Date: 20260601
Title: Growth Curve of *Bacillus subtilis* 168 in normoxia and hypoxia conditions.
Purpose: To test the growth curve of *Bacillus subtilis* 168 in different conditions.
Procedure:

1. Culture the *Bacillus subtilis* 168 in culture tube overnight.
2. Check the OD_{600} (2.04 A).
3. Dilute the *Bacillus subtilis* to 1:25 in 40 mL LB.
4. Culture the cells in 37°C with 200 rpm shaker; different conditions were performed.
5. Measure the OD_{600} per hour.

Results:

Hours	Normoxia (A)	Hypoxia (A)
1	0.175	0.192
2	0.658	0.658
3	1.132	1.045
4	1.461	1.346
5	1.502	1.410
5.75	1.476	1.440

Done by Gabriel.
ELN by Sophia Y.

Date: 20260601
Title: Hydroponics medium
Purpose: Prepare 1/2 MS medium for hydroponics
Procedure:

1. Follow the protocol (for 1000 mL of 1/2 MS medium)

1/2 MS medium powder(without sucrose)	2.3 g
MES buffer	0.5 g
Distilled water	1000 mL

2. Add the contents into a 1000 mL
3. Use a hot plate to stir (with a magnetic stirring rod) until no sediments are left in the solution
4. Use the pH probe and measure the pH of the solution, make sure the pH is 5.7~5.8
5. If the pH is too low, add 0.1M KOH slowly to increase it
6. Wrap the neck of the flask with aluminum foil and stick a piece of indicator tape on top

7. Autoclave for 20 min at 127°C
8. Wait for the pressure to reach 0 before retrieving the flask
9. Add 2.2 mL of medium into each tube rack

Done by Chloe , Sophie C

Date: 20260602
Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:

1. Add 5 µL of 10 ng/µL of Csn:AmilCP and Csn ACCD to 100 µL competent cell
   Add 5 µL of 100 ng/µL of Csn:AmilCP and Csn ACCD to 100 µL competent cell
2. Incubate in ice for 20 min
3. Heatshock of 42 °C for 45 sec
4. Incubate on ice for 3 min
5. Add 1000 µL recovery medium in each microtube
6. Place in incubator 37°C for 48 min
7. Spin down at 5000 g for 5 min
8. Remove 1000 µL and resuspend the cell pellet
9. Spread the mixture on the LB agar plate with ampicillin
10. Place in incubator at 37 °C overnight (16 hr)

Done by Sarah C , Olivia L , Sophie C

Date: 20260602
Title: DNA purification of promoter-PcpcG2-veg & EEBD
Purpose: Isolate and concentrate pure promoter
Procedure:

1. Pipette 100 µL PCR products in each of the four 1.5 mL centrifuge tubes.
2. Add 500 µL of GP buffer to each tube and mix well.
3. Place a GP column into a collection tube → add all the mixture from step 1 into the tube.
4. Load no more than 700 µL into the column each time.
5. Centrifuge 1 min + discard flow through.
6. Add 500 µL WN buffer.
7. Centrifuge 1 min + discard flow through.
8. Add 500 µL WS buffer.
9. Centrifuge 1 min + discard flow through.
10. Centrifuge 8 min at 23444 g and discard flow through.
11. Place the column into a new microcentrifuge tube.
12. Add 30 µL of nuclease free water into the center of the membrane without touching the membrane.
13. Incubate for 3 min and centrifuge at full speed for 2 min to elute DNA.
14. Measure the concentration of DNA and label on the tube. Store DNA at -20°C until use.

Done by Chloe.
ELN by Sophia Y.

Date: 20260602
Title: PCR of PcpcG2-veg & EEBD
Purpose: To magnify the number of desired DNA sequence through replication in the PCR machine
Procedure:

1. Prepare the following master mix.

Reagant	1 rxn (50 µL)	6 rxn (300 µL)
10X ExTaq Buffer	5 µL	30 µL
2.5 mM dNTP mix	4 µL	24 µL
10 µM forward primer	2 µL	6 µL
10 µM reverse primer	1 µL	6 µL
DNA template (10 ng/µL)	1 µL	6 µL
PCR grade water	37.75 µL	226.5 µL
ExTaq polymerase	0.25 µL	1.5 µL

2. Thermal cycle

95℃	5min	1cycle
95℃	30sec	35 cycles
55℃	30sec	
72℃	1min	
72℃	5min	1cycle

Done by Chloe and Sophie C.
Supervised by Gabriel.
ELN by Sophia Y. Sophie H.

Date: 20260603
Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:

1. Add 2 µL of 10 ng/µL of SamyQ: AmilCP, Ho1: pcyA, ccaSm3, ccaR, and Csn:Lea to 30 µL competent cell.
2. Incubate on ice for 20 min.
3. Heatshock at 42 °C for 45 sec.
4. Incubate on ice for 3 min.
5. Spread on the LB agar plate with Ampicillin.
6. Place in an incubator at 37 °C overnight (16 hr).

Done by Sophie L, Sophie C, Sophia Y.
Supervised by Chars.
ELN by Sophia Y.

Date: 20260603
Title: Liquid culture
Purpose:Grow and amplify bacterial cells from single colonies
Procedure:
pTwist-Amp-Csn:AmilCP 1
pTwist-Amp-Csn ACCD 1
pTwist-Amp-Csn AmilCP 2
pTwist-Amp-Csn ACCD 2

1. Prepare TB medium: 12 mL of TB medium and 12 µL of 50mg/mL Ampicillin.
2. Dispend 3 mL of medium in each tube.
3. Inoculate a colony into each tube.
4. Place tubes in incubator at a slanted angle

Done by Sophie L, Sophie C.
Supervised by Chars.
ELN by Sophia Y.

Date: 20260604
Title: Miniprep
Purpose: To isolate pTwise-Amp-High-Csn:AmilCP/ pTwise-CSN-ACCD
Procedure:

1. Transfer 1.8 mL liquid culture to 2 mL microtubes
2. Centrifuge at 16000 g 1 min & discard supernatant
3. Repeat step 1&2 steps until transfer all the liquid culture
4. Add 200 µL of MX1 buffer into each microtube & resuspend pellet
5. Add 250 µL of MX2 buffer and invert 4-6times until solution becomes clear
6. Incubate at room temp for 2 min
7. Add 350 µL of MX3 buffer and invert 4-6 times
8. Centrifuge for 5 min at 23444 g
9. While centrifuging, assemble column collection tube
10. Get 700 µL of the supernatant and transfer it in the column
11. Centrifuge for 30 sec at 7000 g
12. Repeat steps 9 and 10 until the supernatant is completely used up
13. Add 500µL WN buffer & centrifuge for 30 sec at 7000 g
14. Discard the flow-through
15. Add 700µL WS buffer & centrifuge for 30 sec at 7000 g
16. Discard follow through & centrifuge empty column for 3 min at 10000 g to remove residual ethanol
17. Place the column in an empty 1.5 mL microcentrifuge tube and apply 50 µL of ddH_2O to the center of column → rest 3 min & centrifuge for 3 min 10000 g

Result:

Biodrop Results	concentration	A260/230	A260/280
pTwist-Amp-Csn:AmilCP1	255 ng/µL	1.635	2.143
pTwist-Amp-Csn-ACCD1	239 ng/µL	1.691	2.146
pTwist-Amp-Csn:AmilCP2	305 ng/µL	1.640	2.148
pTwist-Amp-Csn-ACCD2	278.0 ng/µL	1.007	2.014

Done by Chloe, Sophie L, Anton, Noah
Supervised by Chars

Date: 20260604
Title: Prepare 0.8% Agarose gel for DNA electrophoresis
Purpose: To prepare a 0.8% agarose gel for analytical DNA electrophoresis.
Procedure:

1. Assemble the mold and tray, and prepare the comb
2. Weight 0.16 g agarose powder and 20 ml 1X TAE solution
3. Add 1X TAE and powder to the conical flask

4. Cover the flask with plastic wrap, use tip to puncture a hole on it
5. Microwave for 30 sec , then take it out and agitate it
6. Microwave for 10 sec and agitate it, repeat this step until there is no particle in melted gel
7. Wait until the melted gel stops smoking but is still hot
8. Add 1 ml safety dye and agitate it
9. Pour the melted gel into the tray, then assemble the comb

Done by Sophie L, Noah
Supervised by Chars
Recorded by Sophie H.

Date: 20260604
Title: Plasmid Linearization by Enzyme Digestion
Purpose: To confirm plasmid size for each samples before moving on to golden gate
Procedure:
A: Csn: AmilCP1
B: Csn: ACCD1
C: Csn:AmilCP2
D: Csn:ACCD 2

1. Calculate the reaction formula as follows:

	A	B	C	D
10X Cutsmart buffer	2 µL	2 µL	2 µL	2 µL
EcoRI-HF	0.5 µL	0.5 µL	0.5 µL	0.5 µL
DNA	2 µL	1.8 µL	1.6 µL	1.8 µL
ddH_2O	15.5 µL	15.7 µL	15.9 µL	15.7 µL

2. Prepare the reaction:
   2-a: Add water, DNA, and buffer
   2-b: Then add enzymes
3. Incubate in MiniAmp thermocycler at 37 °C indefinitely until the laboratory is ready to experiment a gel electrophoresis to size the STK DNA

Done by Sophie L. , Noah
Supervised by Chars
Recorded by Sophie H.

Date: 20260604
Title: Liquid culture
Purpose:grow and amplify bacterial cells from single colonies
Procedure:
A: pTwist-Amp-SamyQ:AmilCP

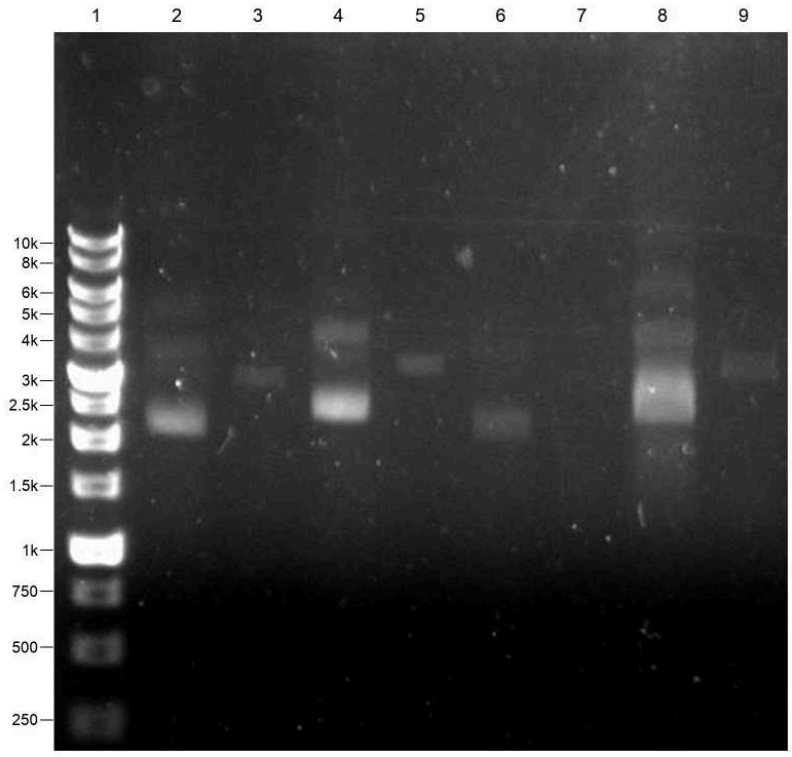
B: pTwist-Amp-Ho1:PcyA
C: pTwist-Amp-CcaRm3
D: pTwist-Amp-CcaR
E: pTwist-Amp-Csn:LEA

1. Prepare TB medium: 35 ml of TB medium and 35 µL of 50 mg/ml Ampicillin
2. Dispense 3 ml of medium in each tube
3. Inoculate a colony into each tube
4. Incubate at 37℃

Done by Sophie L, Sophie C, Abby K, Naomi, Joshua, Sophie H.
Supervised by Chars
Recorded by Sophie H.

Date: 20260604
Title: Electrophoresis
Purpose: Verify the gene synthesis product
Procedure:

1. Load the gel in the layout shown below (5 µL for M; DNA with 1 µL 6x loading dye for all samples):
2. Resolve at 100 v for 30 min



Lane	Sample	Treatment	DNA volume (µL)	DNA Amount(ng)	Size Listed(bp)

1	Omics 1 kb DNA RTU Ladder		5		
2	pTwist-Amp-High-Csn:AmilCP #1	Undigested	5	1200	
3	pTwist-Amp-High-Csn:AmilCP #1	EcoRI digestion, 2 hr	5	125	3051
4	pTwist-Amp-High-Csn:Accd #1	Undigested	5	1400	
5	pTwist-Amp-High-Csn:Accd #1	EcoRI digestion, 2 hr	5	125	3423
6	pTwist-Amp-High-Csn:AmilCP #2	Undigested	5	1500	3051
7	pTwist-Amp-High-Csn:AmilCP #2	EcoRI digestion, 2 hr	5	125	3051
8	pTwist-Amp-High-Csn:Accd #2	Undigested	5	1400	
9	pTwist-Amp-High-Csn:Accd #2	EcoRI digestion, 2 hr	5	125	3423

Done by Chars

Date: 20260605
Title: Miniprep and glycerol stock preparation
Purpose:isolate plasmid and create long term frozen stocks for preservation
Procedure:
A: pTwist-Amp-SamyQ:AmilCP
B: pTwist-Amp-Ho1:PcyA
C: pTwist-Amp-CcaRm3
D: pTwist-Amp-CcaR
E: pTwist-Amp-Csn:LEA

1. Prepare Glycerol stock, formula as follows:

	pTwist-Amp-SamyQ: AmilCP	pTwist-Amp-Ho1: PoyA	pTwist-Amp-ccaSm3	pTwist-Amp-ccaR	pTwist-Amp-csn: Lea
50% Glycerol Stock	500 µL	500 µL	500 µL	500 µL	500 µL
Bacteria Culture Medium	500 µL	500 µL	500 µL	500 µL	500 µL

Then store in -80 deg C.

2. Transfer 1 mL TB medium to 1.5 mL microtube.
3. Centrifuge at 16.000 g for 1 min.
4. Discard the supernatant.
5. Repeat steps 2-4 until collected all medium.
6. Add 200 µL MX1 buffer.
7. Resuspend the pellet by vortexing or pipetting.
8. The cells should completely resuspend.
9. Add 250 µL MX2 buffer.
10. Gently mix the buffer well.
11. Do not vortex. Invert microtube 8-10 times.
12. Incubate the mixture at room temperature for 2 min.
13. Add 350 µL MX3 buffer.
14. Immediately and gently mix the solution. (Do not vortex.)
15. Centrifuge at 16,000 g for 5 min.
16. While centrifuging, assemble the column and collection tube.
17. Transfer the supernatant, around 500 µL (take 800 µL next time), to the column.
18. Centrifuge at 7,000 g for 30 sec.
19. Discard the flow through.
20. Add 500 µL WN buffer.
21. Centrifuge at 7,000 g for 30 sec.
22. Discard the flow through.

23. Add 700 µL WS buffer.
24. Centrifuge at 7,000 g for 30 sec.
25. Discard the flow-through.
26. Centrifuge the empty column at 10,000 g for 3 min to remove residual ethanol.
27. Place the column into a new 1.5 mL microtube.
28. Add 50 µL sterile water in the center of the membrane.
29. Incubate the column at room temperature for 3 min.
30. Centrifuge at 10,000 g for 3 min.

Results:
Recorded from biodrop.

	Concentration	A2601230	A2601280
A	201 µL/mL	1.305	1.990
B	211 µL/mL	1.294	1.991
C	187 µL/mL	1.127	1.948
D	119 µL/mL	0.922	1.859
E	195 µL/mL	1.373	2.010

Done by Sophie L, Noah, Anton.
Supervised by: Gabriel.
ELN by Sophia Y.

Date: 20260605
Title: Prepare 0.8% Agarose Gel
Purpose: For DNA electrophoresis confirming PCR product
Procedure:

1. Assemble the mold and tray, and prepare the comb.
2. Weigh 0.16 g agarose powder and obtain 20 mL 1X TAE solution.
3. Add 1X TAE and powder to the conical flask.
4. Cover the flask with plastic wrap, then use a tip to puncture a hole in the plastic wrap.
5. Microwave for 30 sec. Then take it out and agitate it.
6. Microwave for 10 sec and agitate it. Repeat this step until there is no particle in melted gel.
7. Wait until the melted gel stops smoking but is still hot.
8. Add 1 µL safety dye and agitate it.
9. Pour the melted gel into the tray. Then assemble the comb.

Done by Sophie L
Supervised: Chars, Gabriel
ELN by Sophia Y.

Date: 20260605
Title: LB Medium Preparation
Purpose: Prepare LB Medium
Procedure:

1. Formula as below:

LB broth powder	12.5 g
ddH_2O	500 mL

2. Add one magnetic stirring rod to the flask.
3. Place the flask containing the mixture on the hot plate and stir until dissolved.
4. Remove the stirring rod from the flask.
5. Cover the flask with aluminum foil and apply an indicator tape.
6. Place the flask in an autoclave.

Done by Abby T and Abigail.
Supervised by Chars and Gabriel.
ELN by Sophia Y.

Date: 20260605
Title: Plasmid linearization by enzyme digestion
Purpose: to prepare the plasmid for gel electrophoresis
Procedure:
A: pTWIST-Amp-SamyQ-AmilCP
B: pTWIST-Amp-H01-PcyA
C: pTWIST-Amp-CcaSm3
D: pTWIST-Amp-CcaR
E: pTWIST-Amp-Csn:LEA

1. Calculate reaction formula:

	A	B	C	D	E
10X CutSmart buffer	2 µL	2 µL	2 µL	2 µL	2 µL
EcoR1-HF	0.5 µL	0.5 µL	0.5 µL	0.5 µL	0.5 µL
DNA	2.5 µL	2.4 µL	2.7 µL	4.2 µL	2.6 µL
ddH_2O	15 µL	15.1 µL	14.8 µL	13.3 µL	14.9 µL

2. Prepare the reaction
    - Fist add water DNA and buffer

- Then add enzymes

3. Incubate in MiniAmp thermocycler at 37 °C indefinitely until the laboratory is ready to experiment a gel electrophoresis to size the STK DNA

Done by Chloe

Date: 20260606
Title: Golden Gate Assembly
Purpose: directionally assemble a promoter, RBS, coding sequence, and terminator into a destination vector in a single reaction.
Procedure:

1. Assemble the following:

A: pJUMP29-1A-PcpcG2-Veg-EEBD-m1001-SamyQ-B0015

		Dilute
Vector (pJUMP29-1A)	2.48 µL	1:10
Insert Promoter (PcpcG2-Veg)	1.87 µL	1:100
Insert RBS (EEBD-m1001)	3.28 µL	1:100
Insert CDS (pTwist-Qmp-SamyQ)	3.95 µL	1:10
Insert Terminator (B0015)	3.28 µL	1:10
T4 DNA Ligase Buffer	2 µL	
Sterile Water	1.14 µL	
BsaI-HFv2	1 µL	

B:pJUMP29-1A-523100-EEBD-m1001-SamyQ- B0015

		Dilute
Vector (pJUMP29-1A)	2.48 µL	1:10
Insert Promoter (523100)	4.6 µL	1:100
Insert RBS (EEBD-m1001)	3.28 µL	1:100
Insert CDS (pTwist-Qmp-SamyQ)	3.95 µL	1:10
Insert Terminator (B0015)	3.28 µL	1:10

T4 DNA Ligase Buffer	2.6 µL	
Sterile Water	23 µL	
BsaI-HFv2	1 µL	

2. Incubate both assembly mixtures in a PCR machine using the appropriate Golden Gate Assembly program.

95°C	5 min
95°C	30 sec
55°C	30 sec
72°C	1 min
72°C	5 min

Done by: Chloe, Sophie L., and Ryan Y.
Supervised by: Hachi.

Date: 20260606
Title: *Bacillus subtilis* growth rate in bioreactor
Purpose: test the rate *Bacillus subtilis* growth in bioreactor
Procedure:

1. Measure OD600 *Bacillus subtilis* growth rate in bioreactor and incubator every 30mins

Results:

	Culture in bioreactor	Culture in incubator
0 min	0.172	0.172
30 min	0.108	0.189
60 min	0.101	0.205
90 min	0.194	0.217
120 min	0.130	0.230
150 min	0.178	0.347
180 min	0.166	0.314
210 min	0.112	0.309

240 min	0.123	0.377
270 min	0.136	0.442
300 min	0.126	0.556
330 min	0.125	0.698
360 min	0.083	0.811
390 min	0.131	0.987

Done by Joshua, Chloe, and Ryan W.
ELN by Sophia Y.

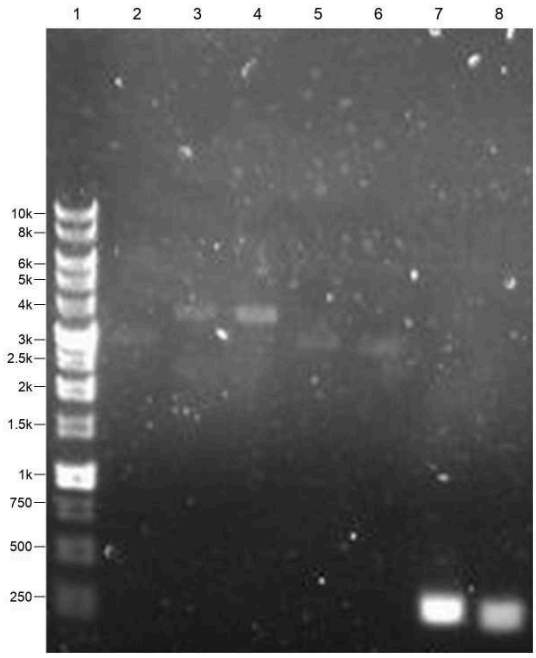
Date: 20260606
Title: Liquid Culture of pJUMP1A–pJUMP29-1D
Purpose: grow and amplify bacterial cells from single colonies
Procedure:

1. Prepare TB with antibiotics (Master Mix): 40 mL TB broth + 40 µL kanamycin (50 mg/mL).
2. Split the master mix into 8 tubes, each containing 5 mL.
3. Inoculate one colony from the glycerol stock into each tube.
4. Place the tubes into the incubator.

Done by: Joshua, Ryan Wei, and Hachi.

Date: 20260606
Title: Electrophoresis
Purpose: Verify the gene synthesis produced.
Procedure:

1. Load the gel in the layout shown below (5 µL for M; 5 µL DNA with 1 µL loading dye for all samples):
2. Resolve at 100 v for 30 min



1: Omics 1kb DNA RTU Ladder 5 uL		
Sample	Plasmid Construct	Expected Size (bp)
2	pTwist-Amp-High-SamyQ:AmilCP 1hr 5 uL (1.005 ug)	
3	pTwist-Amp-High-Ho1:pcyA 1hr 5 uL (1.055 ug)	
4	pTwist-Amp-High-ccaSm3 1hr 5 uL (935 ng)	
5	pTwist-Amp-High-ccaR 1hr 5 uL (595 ng)	
6	pTwist-Amp-High-Csn:Lea 1hr 5 uL (975 ng)	
7	EEBD	
8	Pveg	237

Done by Sophie L, Sophie C, and Sarah C.
Supervised by Hachi.
ELN by Sophia Y.

Date: 2026.06.07
Title:Fresh Weight and Chlorophyll Extraction
Purpose:Measure freshweight and Extract chlorophyll to evaluate arabidopsis growth condition
Procedure:

1. Weigh the fresh weight (fw) of arabidopsis (10 at a time).
2. Take the plant from the agar plate, careful not to hurt it by squeezing the tweezers.
3. Put the 10 plants on a tissue paper, lightly press to remove the agar sticking to the plants.
4. Pick up the plants and put them in a microtube, take care that the roots need to go in, too.
5. Weigh the plants
6. Repeat these steps for as many times as the sets of arabidopsis testing.
7. Place 3 glass bits in each of the microtubes, place in -80°C for 10 minutes.
8. Dilute acetone to 80 %, add 1 mL to each tube.
9. Wait for the chlorophyll to extract, and can use a vortex to crush the plants to speed up the process (done when plants are no longer green.)
10. Add the liquid to the 96 well plate, 200 µL in each well, 3 wells for each sample.
11. Using the ELISA reader, insert the 96 well, run the results(OD645 and OD663) through the equation below.

*Total chlorophyll content (mg/g)
(20.2*OD645+8.02*OD663)V/1000*fw

Results**:**

	Well 1	Well 2	Well 3
E (0.0164)	451.83	479.27	739.02
F (0.0208)	348.56	390.87	391.35
G (0.0207)	346.38	384.54	390.34
H (0.0164)	428.05	473.17	485.98

Done by Phoebe C., Sophia Y., recorded by Phoebe C.
ELN by Sophia Y and Phoebe C.

Date: 2026.06.07
Title: Transformation of BBa_J232119 and Golden Gate Assembly of 6/6
Purpose: To transform competent cells with the Bba_J23119 plasmid and the completed Golden Gate Assembly construct and to isolate cells carrying the desired DNA constructs through antibiotic selection.
Procedure:

1. Turn on the water bath and set at 42 ℃, and prepare the ice.

2. Spread antibiotics on the agar plate.
3. Thaw the competent cell and aliquot 30 µL of the competent cell in each microtube.
4. Add 2 µL of plasmid to the 30 µL competent cell (do not pipette). 2 µL for J23119, 10 µL for J23106 and pepcG2-Veg.
5. Place on ice for 20 minutes.
6. Heat shock at 42 °C for 45 seconds to 1 minute.
7. Place on ice for 3 minutes to recover, add 1 mL SOC medium, and incubate at 37°C for 1 hour.
8. Centrifuge at 5000 g for 5 minutes and remove 1 mL of supernatant, then resuspend pellet.
9. Spread the competent cell on agar plate with kanamycin.
10. Incubate the plate at 37°C for 16 hours.

Done by Chloe, Phoebe, and Sophia Y, recorded by Chloe.
ELN by Sophia Y.

Date: 2026.06.07
Title: Agar Preparation
Purpose: Prepare agar for plant
Procedure:

1. Formula as below:
   2.37 g 1/2 MS Medium
   0.276 g 1/2 MS Medium (no sucrose)
   180 mL ddH_2O
   1.386 g agar
2. Add contents into a beaker.
3. Use the hot plate to stir (with a stirring rod) until no sediments are left in the solution.
4. Adjust the pH with 0.1 M KOH to 5.7-5.8.
   Original pH: 5.40
   Final pH: 5.74
5. Wrap the neck with aluminum foil and place indicator tape on top.
6. Autoclave for 20 min at 127 °C.
7. Wait for the pressure to reach 0 before retrieving the flask.

Done by Sophia Y and Chloe, recorded by Sophia Y.
ELN by Sophia Y.

Date: 20200607
Title: *Bacillus subtilis* growth rate in Bioreater (Continuous from 20260606)
Purpose: continue 20260606 experiment
Procedure:

Time (hr)	Incubator	Bioreactor
24	2.169	0.197
25	2.461	0.342

26	2.402	0.370
27	-	0.304

Sampling from medium : 0.480

Done by Chars

Date: 20260608
Title: Agar Preparation
Purpose: prepare agar for plant growth
Procedure:

1. Formula as below:

1/2 MS medium (with sucrose)	2.37 g
1/2 MS medium (without sucrose)	0.276 g
Agar	1.386 g
ddH_2O	180 mL

2. Add contents into a flask
3. Use the hot plate to stir(with a stirring rod) until no sediments are left inside the solution
4. Adjust the pH using a pH probe to around 5.7 using 0.1M KOH
    a. Record the original pH and final pH values
5. Wrap the neck with aluminum foil and place the indicator tape on top
6. Autoclave for 20 minutes at 127 °C
7. Wait for pressure to reach 0 before opening the autoclave

Done by: Abigail and Sophia Y.
Supervised by Gabriel.
ELN by Sophia Y.

Date: 20260608
Title: Prepare 1.0% Agarose Gel
Purpose: for DNA electrophoresis
Procedure:

1. Assemble the mold and tray, and prepare the comb.
2. Weigh 0.2 g of agarose powder and obtain 20 mL 1X TAE solution
3. Add 1X TAE and agarose powder to the conical flask.
4. Cover the flask with plastic wrap, then use a tip to puncture holes in the plastic wrap
5. Microwave for 30 s and agitate it. Repeat this step until there are no particles in the melted gel.
6. Wait until the gel stops smoking but is still hot.
7. Add 1 µL of safety dye into the gel and agitate it.
8. Pour the melted gel into the tray, then assemble the comb.

Done by Naomi, recorded by Naomi.
Supervised by Gabriel.
ELN by Sophia Y.

Date: 20260608
Title: Colony PCR
Purpose: amplify golden assembly parts with 2 different promoters

Procedure:

1. Prepare the master mix(9 reactions)

The following is the formula for 1 reaction:

Forward Primer(Biobrick prefix)	1 μL
Reverse Primer(Biobrick suffix)	1 μL
Q-Amp Master Mix	10 μL
ddW	8 μL

2. Dispense reaction to 8 strip PCR tube, each 4 colonies for each of the two dishes
3. Pick up colonies and tap 3 times on a separate plate for each of the colonies to incubate
4. Release the tips with the colonies in the PCR tubes and wait for 5 minutes
5. Take out the tips and put tubes in PCR machine
6. Thermal cycle

95 ℃	5 min	1 cycle
95 ℃	30 sec	35 cycles
55 ℃	30 sec	
72 ℃	1 min	
72 ℃	5 min	1 cycle

Done by: Abigail and Sophia Y.
Supervised by: Gabriel.
ELN by Sophia Y.

Date: 20260608
Title: Hydroponic Medium Preparation
Purpose: prepare hydroponic medium for plant growth and testing
Procedure:

1. Measure 2.07 g of 1/2 MS medium in 900mL of de-ionized water
2. Use the hot plate to stir(with a stirring rod) until no sediments are left in the solution
3. Adjust the PH using a plt probe to around 5.7-5.8 use 0.1 M KOH
   Original pH : 4.00
   Final pH: 5.7
4. Split into two beakers (by estimation; equally)
5. Wrap the neck with aluminum foil and place the indicator tape on top
6. Autoclave for 20 min at 127 ℃
7. Wait for the pressure to reach 0 before opening the autoclave

Done by Sophia Y and Abigail
Supervised by Gabriel

Date: 20260608
Title: Sanitizing equipment
Purpose: Disinfect laboratory tools using a 10% bleach solution.
Procedure:

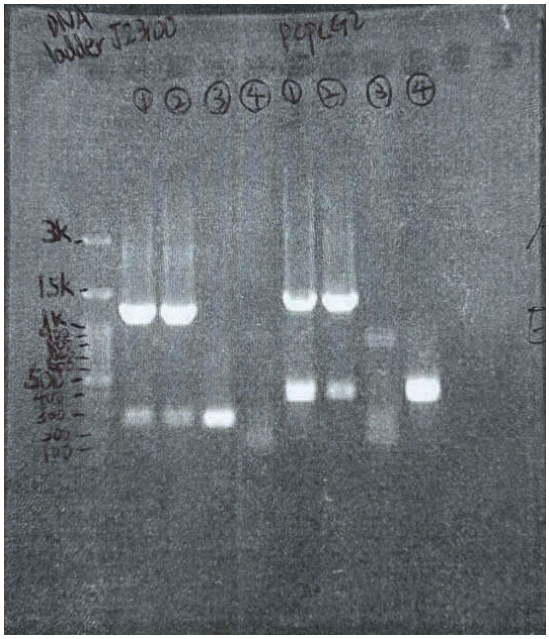
1. Pour 200 mL of pure bleach into a graduated cylinder then into a 2 L beaker
2. Add 1.8 L of dd H_2O to the beaker
3. Pour 10 % bleach into a tub and fully cover the equipment in the tub
4. Wait for 10 minutes
5. Rise the equipment ddH_2O until these is no bleach scent

Done by Abigail and Sophia Y
Supervised by Gabriel

Date: 20260609
Title: Prepare 1.0% Agarose Gel for DNA electrophoresis
Purpose: Do DNA electrophoresis for colony PCR product(6/8)
Procedure:

1. Assemble the mold and tray, prepare the comb
2. Weigh 0.2 g of agarose powder and add it into a flask
3. Add 20 mL of 1xTAE solution into the flask
4. Cover the flask with plastic wrap then use a tip to puncture holes on the plastic wrap
5. Microwave for 30 seconds and agitate it
    a. Repeat this step until all dissolved and no particles left in the mixture
6. Wait until the gel stops smoking but is still hot
7. Add 1 μLof safety dye into the gel and agitate it
8. Pour the melted gel into the tray and then assemble the comb
9. Wait for the gel to cool

Results:



Done by Abby K, Sophia Yeh, Abby T
Supervised by Gabriel

Date: 20260609
Title: Miniprep
Purpose: to amplify P1~4, J1~4
Procedure:

1. Transfer 1 mL TB medium to 1.5 mL microtube
    a. PCPCG2 sample 1
    b. PCPCG2 sample 2
    c. PCPCG2 sample 3

    d. PCPCG2 sample 4
    e. J23100 sample 1
    f. J23100 sample 2
    g. J23100 sample 3
    h. J23100 sample 4
    i. J23119(promoter)

*vector is pJUMP29-1A

2. Centrifuge at 16,000 g for 1 minute
3. Discard the supernatant
4. Repeat steps 1-3 until collected all medium
5. Add 200 µL Mx1 buffer
6. Resuspend the pellet by vortexing or pipetting
7. The cells should completely resuspend
8. Add 250 µL of Mx2 buffer
9. Gently mix buffers well
10. Do not vortex. Invert microtube 8-10 times
11. Incubate the mixture at room temperature for 2 minutes
12. Add 350 µL Mx3 buffer
13. Immediately and gently mix the solution(do not vortex)
14. Centrifuge at 16,000 g for 5 minutes
15. While centrifuging, assemble the column and collecting tube
16. Transfer the supernatant, around 800 microliters to the column
17. Centrifuge at 7,000 g for 30 seconds
18. Discard the flow through
19. Add 500 microliters WN buffer
20. Centrifuge at 7000g for 30 seconds
21. Discard the flow through
22. Add 700 µL of WS buffer
23. Centrifuge at 7,000 g for 30 seconds
24. Discard the flow through
25. Centrifuge the empty column at 10,000 g for 3 min to remove the residual ethanol
26. Place the column into a new 1.5ml microtube
27. Add 50 µL of sterile distilled water on the center of the membrane
28. Incubate the column at room temperature for 3 minutes
29. Centrifuge at 10,000 g for 3 minutes

Done by: Abigail, Abby K, Jacquelyn, Sophia Y
Supervised by Gabriel

Date: 20260609
Title: Enzyme Digestion of J2 (J23100)
Purpose: cut DNA molecules at specific sequence sites
Procedure:

1. Calculate reaction formula

- Double restriction

	J23110(J1)	pNW33N-L2 (sfGFP) (S1)
10X CutSmart buffer	5 µL	5 µL
EcoRI-HF	1 µL	1 µL
PstI	1 µL	1 µL
DNA	9.8 µL	13 µL
ddH_2O	33.2 µL	30 µL

- linearization(single)

	J23110(J3)	J23110(J2)
10X CutSmart buffer	2 µL	2 µL
EcoRI-HF	0.5 µL	0.5 µL
DNA	4.6 µL	4.9 µL
ddH_2O	12.9 µL	12.6 µL

2. Prepare the reaction
    a. First add water, DNA, and buffer
    b. Then add enzyme

(Prepare double restriction twice)

3. Incubate in MiniAmp thermocycler at 37 °C indefinitely until the laboratory is ready to experiment a gel electrophoresis to size the STK DNA

Done by: Chloe, Sophie L, Abby T, Abigail
Supervised by Gabriel

Date: 20260609
Title: Prepare 1.0% agarose gel for electrophoresis
Purpose: prepare gel for electrophoresis
Procedure:

1. Assemble the mold and tray, and prepare the comb
2. Weigh 0.2 g of agarose powder and obtain 20 mL 1X TAE solution
3. Add agarose powder and 1X TAE to a conical flask
4. Cover the flask with plastic wrap , then use a tip to puncture holes in the plastic wrap
5. Microwave for 30 sec. Then take it out and agitate
6. Microwave for 10 sec and agitate it. Repeat this step until there is no particle in melted gel
7. Add 1 µL of safety dye into the gel and agitate

8. Pour the melted gel into the tray. Then assemble the comb

Done by Sophie L
Supervised by Gabriel

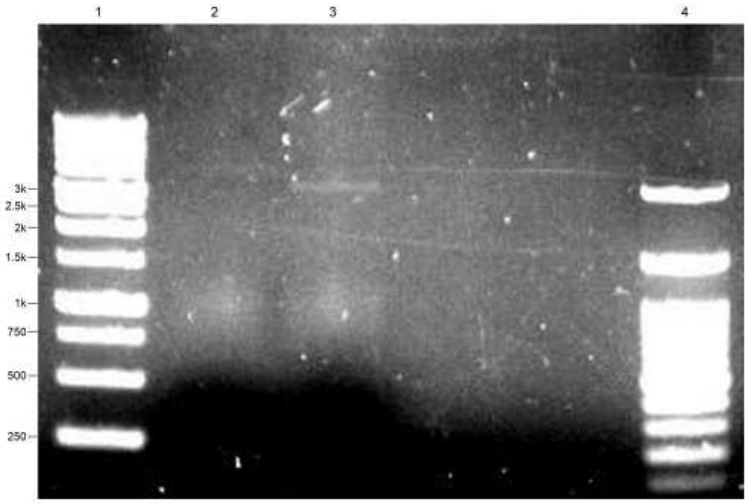
Date: 20260610
Title: Prepare 1.0% Agarose Gel for Electrophrosis
Purpose: Preparation for separating DNA fragments
Procedure:

1. Assemble the mold and tray.
2. Weight 0.2 g of agarose powder and add it into a flask.
3. Add 20 ml of 1x of TAE solution into the flask.
4. Cover the flask with plastic wrap and puncture some holes.
5. Microwave for 30 seconds and agitate it. Repeat this step until all particles are dissolved.
6. Wait until the gel stops smoking but is still hot.
7. Add 1 μL of safety dye into the gel and agitate it.
8. Pour the melted gel into the tray then assemble the comb.

Done by Abby T and Ethan Chang
Supervised by Chars
Recorded by Audrey H.

Date: 20260610
Title: Gel Electrophrosis
Purpose: separate and visualize DNA fragments
Procedure:

1. Load the gel in the layout shown below (5 μL for M; 4 μL DNA with 1 μL loading dye for all samples):
2. Resolve the DNA by 100V 30 minutes.



Lane	Sample	Notes	Volume (μL)
1	Omics 1 kb DNA RTU Ladder	DNA marker	5
2	J23100 sample 2	Highest A230/260	5

3	J23100 sample 3	Highest A230/260	5
4	[DM2300] ExcelBand™ 100 bp+3K DNA Ladder	DNA marker	5

Done by Abby T
Supervised by Chars
Recorded by Audrey H.

Date: 20260610
Title: DNA Ligation
Purpose: Insert the J23100 driven Csn:AmilCP into pNW33N-L2 (sfGFP)
Procedure:

1. Calculate the Insert-Vector Ratio for DNA Ligation based on the molar ratio of 1:3 and formula as below:
2. The ratio as below:

Vector : Insert
1 : 3.58

J23100	3.9 pmol
pNW33N-L2 (sfGFP)	4.3 pmol

3. Prepare the reaction by the following table:

	Volume (μL)
10X T4 DNA Ligase Reaction Buffer	2 μL
pNW33N-L2 (sfGFP)	4.3μL
J23100	3.9μL
Nuclease- Free water	8.8μL
T4 DNA Ligase	1μL

4. Leave for 2 minutes.

Done by Jacquelyn, Abby T and Abigail
Supervised by Chars
Recorded by Audrey H.

Date: 20260610
Title: Electrophoresis and Gel extraction
Purpose: Isolate desired DNA fragment from enzyme digestion
Procedure:

1. Mix 50 µL DNA with 10 µL 6X loading dye.
2. Loading sample by the following layout.

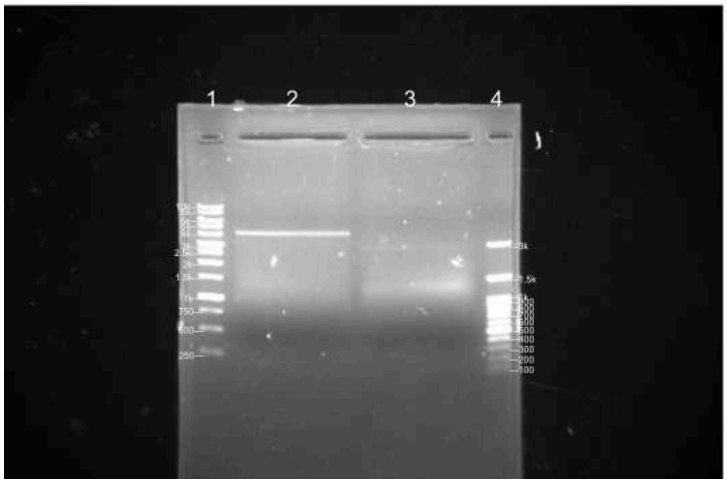
M	A	B	C	
5	60	60	5	volume (µL)

M= 1kb DNA ladder (blue)
A= S1
B= J1 DR
C= 1 kb DNA ladder (green)

3. Resolve the DNA by 100V 30 minutes.

Results**:**

260610 Gel extraction



Lane	Sample	Treatment	Volume(µL)
1	[DM2300] ExcelBand™ 100 bp+3K DNA Ladder		5
2	pNW33N-L2(sfGFP)	EcoRI/PstI digestion, 16 hr	5
3	J23100 Transcript unit	EcoRI/PstI digestion, 16 hr	5

4	Omics 1 kb DNA RTU Ladder		5

Done by Ethan Chang
Supervised by Chars
Recorded by Audrey H.

Date: 20260610
Title: Gel Extraction
Purpose: DNA fragment isolation and purification.
Procedure:

1. Use a sanitized sharp knife to cut the gel band containing the desired DNA fragment under the UV light box.
2. Place each 2 gel bands into separated and labeled 2ml tubes (pNW33N-L2 (sfGFP) EcoRI-PstI and J23100 EcoRI-Pst1).
3. Add 700 µL of GP Buffer into each of the tubes.
4. Place the tubes in a dry bath to dissolve the gel slices, check every 2 minutes until entirely dissolved.
5. Centrifuge for 30 seconds. Discard the flow-through. Repeat this step for the rest of the mixture.
6. Wash the column once with 0.5 ml of WN Buffer by centrifuging for 30 seconds. Discard the flow through.
7. Wash the column once with 0.5 ml of WN Buffer by centrifuging for 30 seconds. Discard the flow through.
8. Centrifuge the column at full speed 23444g for 3 minutes to remove residual ethanol.
9. Place the column onto a new centrifuge tube. Add 15 µL of Nucleus free water that is heated by the thermal shaker onto the center of the membrane.
10. Stand the column for 2 minutes and centrifuge at full speed 23444g for 2 minutes to elute DNA.
11. Turn on BioDrop, adjust the setting to pathlength unit (concentration).
12. Clean the sample port with 1 µL of Nucleus free water.
13. Calibrate.
14. Load 1 µL of DNA from the 2 tubes each time and record down the results.
15. Clean the port with Nucleus Free water and close BioDrop.

Done by Abby T
Supervised by Chars
Recorded by Audrey H

Date: 20260610
Title: Transformation of pNW33N-L2 (sfGFP)
Purpose: Bacterial transformation with plasmid DNA.
Procedure:

1. Turn on the water bath and set it to 42 °C and prepare ice.
2. Take out agar plate with antibiotics (Chloramphenicol).

3. Thaw 100 µL of the competent cell and add 10 µL plasmid (don’t pipette).
4. Place on ice for 10 minutes.
5. Heat shock at 42 °C for 45 to 60 seconds.
6. Place on the ice for 5 minutes to recover and add 1 mL SOC medium, then incubate for 1 hour at 37 °C.
7. Centrifuge at 16,000 g for 30 seconds, remove 1 mL supernatant and resuspend pellet.
8. Spread competent cells on agar plate with Chloramphenicol .
9. Incubate at 37 °C for 16 hours.

Done by Jacquelyn, Abby T, Chloe, Abigail, Sophie L, and Olivia L
Supervised by Chars
Recorded by Audrey H

Date: 20260611
Title: Agar Prep
Purpose: make 1% sucrose agar plates in rectangular form (2 plates)
Procedure:

1. Formula as below:
    - sucrose: 1.0 g
    - w/o sucrose 1/2 MS medium: 0.23 g
    - agar: 1.0 g
    - MES: 0.05 g
    - ddH_2O: 100 mL
2. Add contents into a beaker with ddH_2O
3. Stir with a stirring rod on a hot plate until no sediments are left in the solution.
4. Add KOH & HCl to adjust the pH to 5.7 ~ 5.8.
    - original pH: 4.1
    - final pH: 5.7
5. Cover flask with aluminum foil & place indicator tape on top.
6. Autoclave for 20 minutes at 127 °C.
7. Pour the agar mixture into the rectangular plates and wait for it to solidify in the BSC

Done by: Abigail, Jacquelyn, Sophie C, Joshua
Supervised by: Chris

Date: 20260612
Title: Transformation PcpcG2-veg/EEBD-MF001
Purpose: Bacterial transformation with plasmid DNA.
Procedure:

1. Turn on the water bath, set it to 42 ℃ and prepare the ice.
2. Mix 100 µL competent cell with 2 µL plasmid (don't pipette)
3. Incubate on ice for 20 minutes.
4. Heat shock at 42 ℃ for 45 seconds.
5. Incubate on ice for 3 minutes.
6. Put in incubator at 37 °C for 1 hr.
7. Centrifuge 5000 g for 5 minutes.
8. Taking out 1000 µL and resuspend the cell.
9. Spread on the agar plate with Ampicillin.
10. Place it in incubator for 37 °C overnight.

Done by Sarah C.
Supervised by Hachi
Recorded by Sophie Chen, Audrey H

Date: 20260612
Title: TA cloning
Purpose: Create plasmid by cloning insert into vector
Procedure:

1. Calculate the insert volume.
   Ratio: 3:1
   - PcpcG2-veg : 0.01363(µL) / 0.168(µg/mL)=0.072
   - EEBD-MF001:0.01204/0.109=0.125
2. Dilute the insert PcpcG2-veg 20X and EEBD-MF001 10X

Reagents	Standard Reaction (ML)	
2X Rapid ligation Buffer, T4 DNA Ligase	5	5
pGEM-T Easy Vector (50ng) (3015bp)	1	1
T4 DNA Ligase	1	1
Denionized water to final volume of	10	10
insert PcpcG2-veg (242 bp)	1.44	
insert EEBD-MF001 (274 bp)		1.25

3. Mix the reaction by pipetting.
4. Incubate the reaction one hour at room temperature.

Done by Sarah C.

Supervised by Gabriel
Recorded by Audrey H

Date: 20260613
Title: Colony PCR for PCPC62-veg / EEBD-MF001's transformation 0612
Purpose: verify if the bacterial colonies successfully received the PCPC62-veg or EEBD-MF001 transformation inserts.
Procedure:

1. Draw a grid on new LB place with Ampicillin and prewarm in 37 °C (4 x 4 grid)
2. Prepare PCR reaction master mix in 1.5 mL microtube

2x Taq master mix	180 µL
Prefix primer	18 µL
Suffix primer	18 µL
PCR- grade water	144 µL

3. Aliquot mix to two 8-strip PCR tubes
4. Label the colonies numerically 1 to 8 (1 for EEBD & 1 for PCP)
5. Touch the single, well isolated colony gently using a sterile pipette tip
6. Streak to the labelled grid & Eject the tip into labelled PCR tube
7. Wait for 2 minutes and remove the tips and place reaction in PCR machine for the following times

Cycle	Step	Temperature (°C)	Duration
1	Denature	95	5 minutes
35	1. Denature 2. Annealling 3. Elongation	1. 95 2. 55 3. 72	4. 30 seconds 5. 40 seconds 6. 1 minute/Kb
1	Elongation	72	5 minutes

Done by Olivia Lin
Recorded by Audrey H

Date: 20260613
Title: LB Medium Reparation
Purpose: prepare LB medium
Procedure:

1. Formula as below (mixture for each of 2 flasks):

LB Broth Powder	12.5 g
ddH_2O	500 ml

2. Add one magnetic stirring bar to the flask
3. Place the flask containing the mixture on the hot plate and stir until dissolved
4. Remove the stirring bar from the flask
5. Cover the flask with aluminum foil and apply the indicator tape
6. Autoclave for 20 minutes at 127 °C

Done by Abby T, Sara Chen

Supervised by Chars

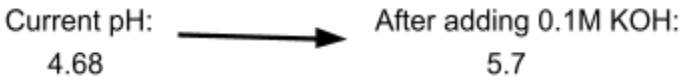
Recorded by Audrey H

Date: 20260614
Title: 1/2 MS medium 1% agar prep
Purpose: Arabidopsis salinity growth test
Procedure:

1. Prepare the following in a flask:

1/2 MS medium powder	1.15 g
MES buffer	0.25 g
Agar powder	5 g
Sucrose	5 g
Distilled water	500 ml

2. Add half of the water in and use a hot plate to stir until no more sediment
3. Add the remaining water and stir
4. Make sure the pH is between 5.6 to 5.7



5. Transfer 250 ml of the mixture into another flask.
6. Add 1.461 g of NaCl into a flask
    a. Label each flask clearly
7. Separate both the NaCl and control mixtures into 4 separate flasks.
8. Label clearly each flask and their respective trehalose concentrations
    a. 0 mM, 0.25 mM, 0.5 mM, and 1 mM
9. Add their respective trehalose amounts:

Concentration	100 mM trehalose
0 mM	0 µL
0.25 mM	125 µL
0.5 mM	250 µL
1 mM	500 µL

10. Cover with aluminium foil and label with indicator tape.
11. Autoclave at 127 °C for 20 minutes.
12. Pour into square petri dishes under the BSC (around 50ml each) using a pipette gun
13. Let solidify under room temperature

Done by Abby K, Abigial

Recorded by Audrey H

Date: 20260614
Title: Bioreactor growth test
Purpose: measure the *Bacillus subtilis* 169 grow in the bioreactor
Procedure:

1. Assemble the bioreactor
2. Add 20 mL on culture medium to bioreactor & flask (volume 500mL)
3. Sample and measure OD600 culture in 37℃ , 200rpm orbital shaker (flask) and 200 rpm stir (bioreactor)

Hr	Time	Flask	Bioreactor
0	11:00	0.122	0.162
0.25	11:15	0.130	0.148
0.5	11:42	0.154	0.137
0.75	12:02	0.173	0.176
1	12:19	0.226	0.262
	12:36	0.226	0.391
	12:53	0.257	0.517
	13:10	0.298	0.767
2	13:25	0.380	0.895
	13:46	0.408	0.784
	2:05	0.439	0.831
	14:22	0.583	0.961
	14:37	0.537	0.875
	14:54	0.606	0.938
	15:10	0.640	0.879
	15:28	0.761	0.883
	15:43	0.832	0.819
	16:00	0.634	0.841
	16:19	0.622	0.741

	16:41	0.572	0.764
	16:58	0.686	0.916
	17:15	0.674	0.826
	17:32	0.657	0.812
	17:51	0.817	0.817
	18:10	0.732	0.843

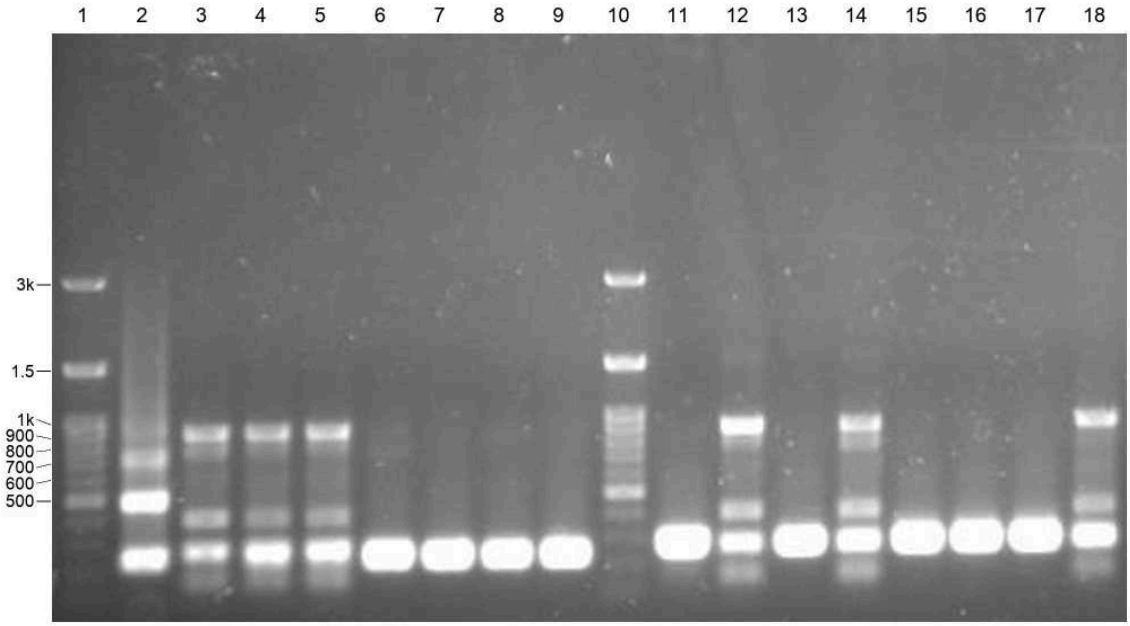
Date: 20260614
Title: Make 1% Agarose gel
Purpose: prepare gel for electrophoresis
Procedure:

1. Assemble mold+tray and prepare the comb
2. Mix 0.4 g agarose and 40 mL 1XTAE solution in a conical flask
3. Cover flask with plastic wrap and puncture holes
4. Microwave in short periods(30 sec then 10 sec) until powder is all melted
5. Add 2 μL of safety dye into mixture and agitate
6. Pour mixture into the tray and assemble comb

Done by Joshua W+Phoebe

Date: 20260614
Title: Gel electrophoresis for PcpcG2-veg and EEBD-MF001
Purpose: test colony PCR result
Procedure:

1. Loaded (Colony PCR product of 20260613 Done by Olivia) with 1~8 ladder (PcpcG2-veg) and 9~16 with EEBD-MF001



Lane	Sample	Volume(μL)
1	[DM2300] ExcelBand™ 100 bp+3K DNA Ladder	5
2	PCPCG2-veg sample 1	5
3	PCPCG2-veg sample 2	5
4	PCPCG2-veg sample 3	5
5	PCPCG2-veg sample 4	5
6	PCPCG2-veg sample 5	5
7	PCPCG2-veg sample 6	5
8	PCPCG2-veg sample 7	5
9	PCPCG2-veg sample 8	5
10	[DM2300] ExcelBand™ 100 bp+3K DNA Ladder	5
11	EEBD-MF001 sample 1	5
12	EEBD-MF001 sample 2	5
13	EEBD-MF001 sample 3	5
14	EEBD-MF001 sample 4	5

15	EEBD-MF001 sample 5	5
16	EEBD-MF001 sample 6	5
17	EEBD-MF001 sample 7	5
18	EEBD-MF001 sample 8	5

Done by Joshua

Date: 20260615
Title: Miniprep
Purpose: Amplifying pcpcG2-veg and EEBD-MP001
Procedure:
A,B: PcpcG2-veg
C,D: EEBD-HF001

1. Transfer 1mL TB medium to 1.5 mL microtube
2. Centrifuge at 16000g for 1 min
3. Discard the supernatant
4. Repeat steps 1-3until collected all medium
5. Add 200μL MX1 buffer
6. Resuspend the pellet through vortexing or pipetting
7. The cells should be completely reasuspend
8. Add 250μL of MX2 buffer
9. Gently mix the buffer well by inverting the microtube 8-10 times
10. Incubate the mixtures at RT for 2 min
11. Add 350μL MX3 buffer
12. Immediately and gently mix through invert 4-5 times
13. Centrifuge at 16000 g for 5 min
14. While centrifuging assemble the column and collection tube
15. Transfer the supernatant, around 800μL to the column
16. Centrifuge at 7000 g for 30 sec
17. Discard the flow-through
18. Add 500μL WN buffer
19. Centrifuge at 7000 g for 30 sec
20. Discard the flow-through
21. Add 700μL WS buffer
22. Centrifuge at 7000 g for 30 sec
23. Discord the flow-through
24. Centrifuge the empty column at 10000 g for 3 min to remove ethanol
25. Place the column into a new 1.5 mL microtube
26. Add 50μL sterile distilled water on the center of the membrane
27. Incubate the column at RT for 3 min
28. Centrifuge at 10000 g for 3 min

Results:
The data below was measured with BioDrop.

	Concentration (ng/μL)	A260/280
A	93	1.898
B	90	1.957

C	168	1.953
D	74	1.897

Done by Abigail, Abby T, Abby K, and Audrey Hsieh.
Supervised by Gabriel Ng

Date: 20260615
Title: 1/2 MS Medium Preparation 700 mL
Purpose: Arabidopsis planting
Procedure:

1. Prepare the following in a large beaker
    a. 1/2 MS medium powder (w/o sucrose) : 1.61 g
    b. MES buffer: 0.35 g
    c. Distilled water: 700 mL
    d. Sucrose (1%): 7 g
    e. Agar powder (1%): 7 g
2. In two separate flasks, put 3.5g of agar powder in broth
3. Mix the contents in the beaker using & hot plate until no sediments are left.
4. Adjust the pH until it is between 5.7-5.8 -> add 0.1M KOH to increase pH: before:3.92, after: 5.72
5. Pour 350mL of the mixture into each of the two flasks with agar powder
6. Use a stirring bar to stir until no sediments left
7. Cover with aluminium foil and stick indicator tape on top
8. Autoclave at 127℃ for 20 minutes
9. Under the biosafety cabinet, use a pipette gen to separate the medium into 12 square petri dishes
10. Let the petri dishes solidify at room temperature

Done by Ethan Chang, Abby K, Sophia Yeh, Abby T, Audrey Hsieh

Date: 20260617
Title: 1/2 MS hydroponic box medium
Purpose: Make a total of 4 liters of hydroponic medium for hydroponic box
Procedure:

1. Measure 4.6g 1/2 MS medium(without sucrose) and 1g of MES in 2L ddH_2O water
    a. Measuring formula ⇒ (2.3g/L)*2 = 4.6
2. Use the hot plate to stir(with a stirring rod) until no sediments are left in the solution
3. Determine pH of the solution and adjust the pH using a pH probe to around 5.6-5.7 using 1M KOH
    a. Record original and final pH
4. Wrap the neck with aluminum foil and place the indicator tape on top
5. Autoclave at 127°C for 20 minutes
6. Wait for the pressure to reach 0 before opening the autoclave

Done by Abigail, Reneee
Supervised by Chars

Date: 20260617
Title: LB Agar
Purpose: Prepare LB Agar for further bacteria cultivation and transformation
Procedure:

1. Measure 6.25g of LB broth and 3.7g of agar in 250ml of distilled water
2. Use the hot plate to stir until no sediments remain in the solution
3. Autoclave at 127°C for 20 min
4. Bring to BSC to add 250µL of Chloramphenicol into the autoclaved agar solution
5. Pipette the solution (10-15µL) into respective petri dishes and leave to dry (solidify)

Done by Abigail, Renee
Supervised by Chars
Record by Sophie H

Date: 20260617
Title: Transformation of pTwist-Amp-HIgh SamyQ:TAT-HA-AmilCP
Purpose: To transform engineered
Procedure:

1. Add 15 µL of bacteria and 1µL of plasmid in the microtube (don't pipette)
2. Place on ice for 20min
3. Heat shock at 42°C for 45sec
4. Place on the ice for 3min to recover
5. Spread the mixture onto the LBA plate
6. Place in the incubator at 37°C overnight

Done by Abigail
Supervised by Chars
Record by Sophie H.

Date: 20260618
Title: Miniprep
Purpose: Collecting miniprep from 20 STK kit parts
Procedure:

1. Label the following:

	name	concentration(ng/μL)	A260/230	A260/280
A1	pSTK-1A-sfGFP	115.0	1.264	1.917
A2	pSTK-1B-sfGFP	151.0	1.027	1.888
A3	pSTK-1C-sfGFP	124.0	1.246	1.919
A4	pSTK-1D-sfGFP	132.0	1.179	1.913
A9	pSTK-Exp-1-sfGFP	162.0	1.141	1.884
C10	pSTK-Exp-2-sfGFP	103.0	1.493	1.981
D2	pSTK-0A-Pveg(BBa)	280.0	1.728	2.059
D3	pSTK-0A-PlepA	155.0	1.422	1.987
D4	pSTK-0A-PliaG	130.0	1.150	1.940
D5	pSTK-0A-PxylA	168.0	1.135	1.909
D7	pSTK-0A-Phyperspar	125.0	1.562	1.953
D8	pSTK-0A-BBa_Rodos	78.0 *	1.345	1.902
D9	pSTK-0a-BBa_R7800	122.0	1.173	1.902
D10	psTK-0AB-PtagC + RBS	125.0	1.645	1.984
D11	pSTK-0B-3P_promoer	106.0	1.093	1.828
E7	pSTK-B-RBS0	188.0	1.635	2.066
E10	pSTK-RS-TT-RS-Rib	150.0	1.210	1.974
G11	pSTK-0D-B0015	107.0	1.216	1.981
G12	pSTK-0D-L3S2P21	170.0	0.762	1.789
E2	pSTK-0C-eGFP_codon	142.0	1.406	1.972

2. Transfer 1.8 mL of TB medium to a 2 mL microtubule.
3. Centrifuge for 60 seconds at 16,000 g.
4. Discard the supernatant.

5. Repeat steps 2-3 until there is no TB medium left in the tube
6. Add 200 µl of MX1 buffer.
7. Resuspend the pellet through vortexing or pipetting.
8. The cells should be completely resuspended before moving to step 9.
9. Add 250 µl of MX2 buffer.
10. Gently mix the buffer well by inverting the microtubule back and forth for 8-10 times.
11. Incubate the mixture at room temperature for 120 seconds.
12. Add 350 µl of MX3 buffer to the mixture.
13. Immediately and gently mix via inversion 4-5 times.
14. Centrifuge the mixture at max rcf [23,444 g] for 480 seconds.
15. While centrifuging, assemble the column and collection tube.
16. Transfer the supernatant (around 800 µl) to the column using a 200 µl micropipette to avoid transferring/contaminating the supernatant with the pellet.
17. Centrifuge the column at 7,000 g for 30 seconds.
18. Discard the flow-through liquid in the collection tube.
19. Add 500 µL WN buffer.
20. Centrifuge at 7,000 g for 30 seconds.
21. Discard the flow-through.
22. Add 700 µL WS buffer.
23. Centrifuge at 7,000 g for 30 seconds.
24. Discard the flow-through.
25. Centrifuge the empty column at 10,000 g for 3 minutes (next time 8 minutes,) to remove residual ethanol.
26. Place the column into a new 1.5 mL microtube.
27. Add 50 µL sterile distilled water on the center of the membrane.
28. Incubate the column at room temperature for 3 minutes.
29. Centrifuge at 10,000 g for 3 minutes.

Done by Audrey Hsieh, Sophia Yeh, Abby Kao, Ethan Chang
Supervised by: Chars Hsieh and Gabriel Ng
ELN by Ethan Chang

Date: 20260618
Title: Enzyme Digestion
Purpose: To test if enzymes have a reaction with the newly-miniprepped DNA.
Procedure:

1. Prepare the mastermix:

10X CutSmart (buffer)	40 µl
EcoRI	10 µl
ddH_2O	200 µl

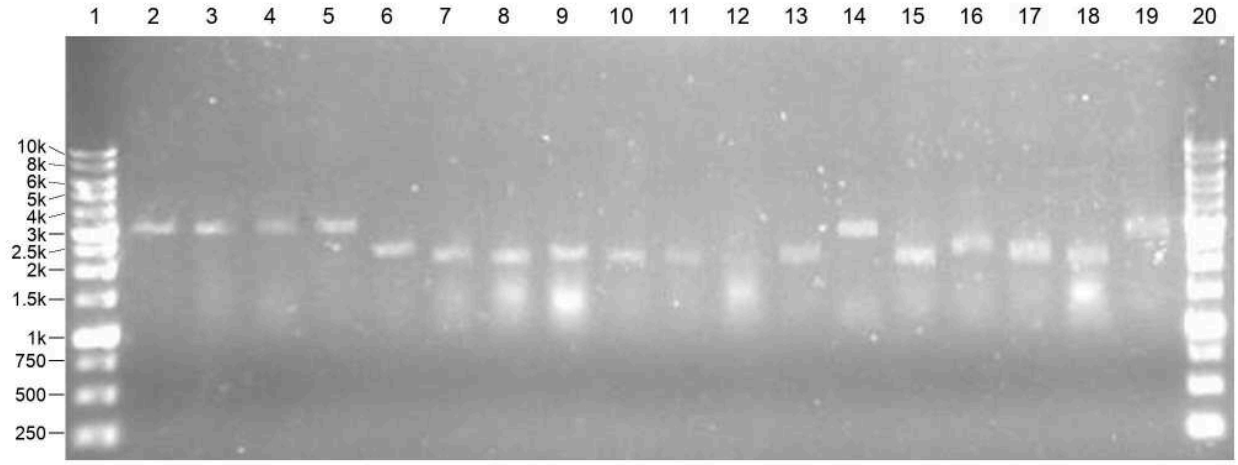
2. Place the contents into a 1.5 mL microtube (EcoRI last).

3. Mix the contents via pipetting, and distribute 15 µl into 18 separate 0.5 mL microtube
4. Add 5 µl of DNA to the 0.5 mL microtube *[from 20260618 Miniprep]*
    a. Add 5 µl of DNA to the tube and label it respectively to its STK name (pSTK…).
    b. Repeat step 4a until all DNA has been added **except** C10 and A9 (Exp-2 sfGFP and Exp-1 sfGFP).
5. Incubate the labeled 0.5 mL microtubes in MiniAmp thermocycler at 37°C indefinitely until the laboratory is ready to experiment a gel electrophoresis to size the STK DNA

Done by Sophia Yeh and Ethan Chang
Supervised by Gabriel Ng
ELN by Ethan Chang

Date: 20260618
Title: Electrophoresis
Purpose: Size the STK DNA
Procedure:

1. Load the gel in the layout shown below (5 µL for M; 5 µL DNA with 1 µL loading dye for all samples):
2. Resolve at 100 v for 30 min



1. Omics 1kb DNA RTU Ladder; 20. Omics 1kb DNA RTU Ladder	
Sample	Plasmid Construct
2	pSTK-1A-sfGFP
3	pSTK-1B-sfGFP
4	pSTK-1C-sfGFP
5	pSTK-1D-sfGFP

6	pSTK-0A-Pveg(BBa)
7	pSTK-0A-PlepA
8	pSTK-0A-PliaG
9	pSTK-0A-PxylA
10	pSTK-0A-Phyperspar
11	pSTK-0A-BBa K0905
12	pSTK-0A-BBa K7800
13	pStk-OAB-PtagC+RBS
14	pSTK-0B-3P promoter
15	pSTK0-B-RBS0
16	pSTK0B-RS-TT-RS-Rib
17	pSTK-0D-B0015
18	pSTK-0D-L3S2P21
19	pSTK-0C-eCFP codon

Done by Abby Kao & Abby Tsai
Supervised by Hachi Wu

Date: 20260618
Title: making agarose gel 1%
Purpose: make 24 well agarose gel for gel electrophoresis
Procedure:

1. Prepare 0.4g of agarose powder and 40mL of 1X TAE in flask
2. Cover with piece of plastic wrap and use a pipette tip to poke 3-5 holes
3. Microwave per 30sec
4. Take it out to agitate
5. Microwave for 5 sec
6. Take out to agitate
7. Repeat steps 4-5 until all powder is dissolved
8. Remove plastic wrap and wait until warm but not too hot to touch
9. Add 2μL of safety dye, pipetting to mix
10. Pour into the mold with 24 well comb
11. Use a pipette tip to press out air bubble

12. Leave to solidify at RT

Done by Abby K and Audrey

Date: 20260619
Title: PCR
Purpose: Amplify arrived DNA parts.
Procedure:

1. Centrifuge primer tubes for 16000 g 1 minute.
2. Add nucleus free water base on the corresponding data sheet to dissolve the primer.
3. Dilute 10x (3 µL primer + 27 µL nucleus free water) each to a new tube.
4. Prepare the following:

Reagent	1 rxn (50 µL)	24 rxn (1200 µL)
10X ExTaq buffer	5 µL	120 µL
25 mM dNTP mix	4 µL	96 µL
10 µM forward primer	1 µL	24 µL
10 µM reverse primer	1 µL	24 µL
DNA temp	1 µL	24 µL
PCR grade water	37.75 µL	906 µL
ExTaq polymerase	0.25 µL	6 µL

5. These are the following templates:

	Name	Concentration (ng/µL)
A	ccaR	119
B	ccaSm3	187
C	Csn: AmilCP	305
D	Csn: ACCD	278
E	Csn: LEA	195
F	Ho1-pcyA	211
G	PcpcG2-veg	90
H	Pveg (STK → Moco)	280

6. Distribute 47 µL of 24 rxn mixture into each tube.
7. Add DNA template and their corresponding primers.

8. Place into PCR machine for the thermal cycle as below:

Thermal cycle number	Temperature (°C)	Time (sec)
1	98	30
2	98	10
3	53	30
4	72	30
5	72	120
6	25	0

Repeat thermal cycles 2, 3, and 4 for 25 times.

Done by Abby T, Abigail, Abby K, Audrey H, and Sarah.
ELN by Sophia Y.

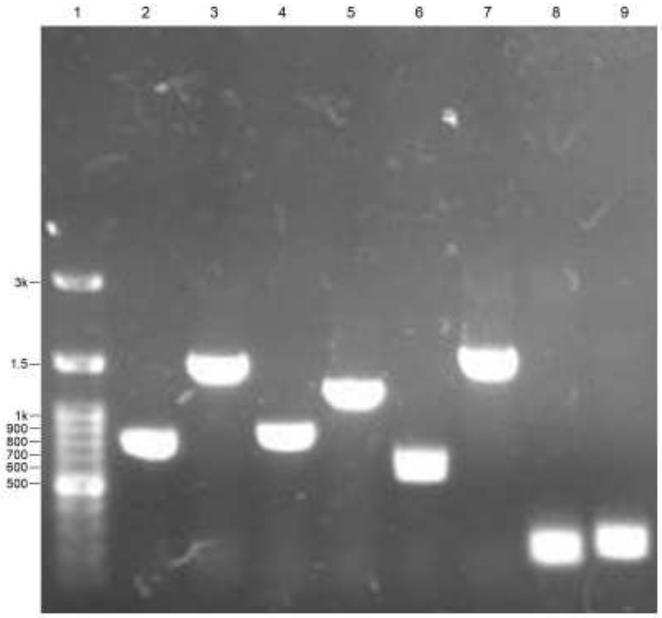
Date: 20260619
Title: Making 1% agarose gel.
Purpose: Make 24 well agarose gel for gel electrophoresis.
Procedure:

1. Prepare 0.4 g of agarose powder and 40 mL of 1X TAE in a flask.
2. Cover with a piece of plastic wrap and use a pipette tip to poke 3-5 holes.
3. Microwave for 30 sec then agitate it. Repeat this step until there's no sediments in the melted gel.
4. Remove the plastic wrap and let the gel cool down until warm but not too hot to touch.
5. Add 2 µL of safety dye into the gel and agitate it.
6. Pour into the assembled rectangular mold with 24 wells.
7. Use a pipette tip to press out bubbles.
8. Leave to solidify at room temperature.

Done by Abby K, Abby T, and Abigail.
ELN by Sophia Y.

Date: 20260619
Title: Electrophoresis
Purpose: Verify the size of the PCR product.
Procedure:

1. Load the gel in the layout shown below (5 µL for M; 5 µL DNA with 1 µL loading dye for all samples):
1. Resolve the samples at 100 V for 30 minutes.



Lane	Sample	Volume(μL)
1	[DM2300] ExcelBand™ 100 bp+3K DNA Ladder	5
2	CcaR	5
3	CcaS	5
4	Csn:AmilCP	5
5	Csn:AccD	5
6	Csn:Lea	5
7	Ho1-PcyA	5
8	PcpcG2-veg	5
9	Pveg (STK→Moco)	5

Done by Abby T and Abby K.
ELN by Sophia Y.

Date: 20260620
Title: Cleaning up
Purpose: To isolate target cell from unwanted cell
Procedure:

1. Prepare 8 1.5 mL centrifuge tubes and add in 750 µL of GB buffer with 150 µL of PCR product; mix well.
2. Place a 6B column on a collection tube; add 500 µL of mixture in.
3. Centrifuge at 10,000 g for 30 seconds. Discard flow through.
4. Repeat steps 2 and three.
5. Add 0.5 mL WN Buffer and centrifuge for 30 seconds; discard flow through.
6. Add 0.5 mL WS Buffer and centrifuge for 30 seconds; discard flow through.
7. Centrifuge at full speed for 3 minutes to remove ethanol residue.
8. Place the column on a new 1.5 mL centrifuge tube and add 30 µL of Elution Buffer.
9. Store the column for 3 minutes and centrifuge at full speed for 2 minutes.
10. Store DNA at 4°C or -20 °C.

Done by Ryan Y and Olivia.
ELN by Sophia Y.

Date: 20260620
Title: 3 liters 1/2 MS Medium
Purpose: Prepare medium for hydroponic system.
Procedure:

1. Prepare the following:
    a. 1/2 MS Medium Powder 3.45 g x 2
    b. MES 0.75 g x 2
    c. RO Water 3 L
2. Have 2 2L beakers filled with 1.5 L of water (measuring cylinder).
3. Add in 3.45 g 1/2 MS medium powder & 0.75 g MES in both beakers; mix well.
4. Tune the pH with KOH until it reaches 5.7-5.8.

	Original pH	Final pH
1st beaker	4.20	5.70
2nd beaker	4.26	5.70

5. Cover the beakers with aluminum foil and autoclave for 20 minutes at 127 °C.

Done by Ryan Y.
ELN by Sophia Y.

Date: 20260620
Title: 10 mL 1/2 MS Agar
Purpose: For hydroponics systems.

Procedure:

1. Prepare:
    a. Agar powder 0.1 g
    b. Sucrose 0.1 g
    c. MES 0.005 g
    d. 1/2 MS Medium 0.023 g
    e. RO water 10 mL
2. Mix all in a beaker and mix well.
3. Tune the pH with KOH until reach 5.7-5.8.
    a. Original pH: 4.8
    b. Final pH: 5.71
4. Cover the beaker and autoclave for 20 minutes at 127 °C.

Done by Ryan Y.
ELN by Sophia Y.

Date: 20260620
Title: Miniprep
Purpose: Extracting plasmid dna from STK
Procedure:

1. Label the following:

Label	Part Name	Concentration (µg/ml)	A260/280
C1	pSTK-0-sfGFP	442.0	2.167
C2	pSTK-1A-sfGFP	788.0	2.153
C3	pSTK-1B-sfGFP	310.0	2.168
C4	pSTK-1C-sfGFP	305.0	2.133
C5	pSTK-1D-sfGFP	369.0	2.133
E4	pSTK-0B-TM_RBS-2	353.0	2.139
E5	pSTK-0B-TM_RBS4	560.0	2.154
E6	pSTK-0B-PET_RBS	344.0	2.150
E8	pSTK0-B-RBS2	400.0	2.126
E9	pSTK0B-RiboJ-RBS0	355.0	2.126

2. Transfer 1.8 mL of TB medium to a 2 mL microtube
3. Centrifuge for 60 seconds at 16,000 g
4. Discard the supernatant
5. Repeat steps 2-3 until there is no TB left in the tube

6. Add 200 µL of MX1 buffer
7. Resuspend the pellet through vortexing or pipetting
8. The cells should be completely resuspended before moving to step 9.
9. Add 250 µl of MX2 buffer.
10. Gently mix the buffer well by inverting the microtubule back and forth for 8-10 times.
11. Incubate the mixture at room temperature for 120 seconds.
12. Add 350 µl of MX3 buffer to the mixture.
13. Immediately and gently mix via inversion 4-5 times.
14. Centrifuge the mixture at max rcf [23,444 g] for 480 seconds.
15. While centrifuging, assemble the column and collection tube.
16. Transfer the supernatant (around 800 µl) to the column using a 200 µl micropipette to avoid transferring/contaminating the supernatant with the pellet.
17. Centrifuge the column at 7,000 g for 30 seconds.
18. Discard the flow-through liquid in the collection tube.
19. Add 500 µL WN buffer.
20. Centrifuge at 7,000 g for 30 seconds.
21. Discard the flow-through.
22. Add 700 µL WS buffer.
23. Centrifuge at 7,000 g for 30 seconds.
24. Discard the flow-through.
25. Centrifuge the empty column at 10,000 g for 3 minutes (next time 8 minutes,) to remove residual ethanol.
26. Place the column into a new 1.5 mL microtube.
27. Add 50 µL sterile distilled water on the center of the membrane.
28. Incubate the column at room temperature for 3 minutes.
29. Centrifuge at 10,000 g for 3 minutes.

Done by Phoebe Chen & Ethan Chang
Supervised by Gabriel Ng
ELN by Ethan Chang

Date: 20260620
TItle: Enzyme Digestion
Purpose: Testing enzymatic reaction with newly-miniprepped DNA
Procedure:

1. Prepare the mastermix:

Materials	Volume
10X CutSmart	20 µL
EcoRI	5 µL
ddH_2O	100 µL

2. Place the contents into a 1.5 mL microtube (EcoRI last)

3. Mix the contents via pipetting, and distribute 15 µL into 10 separate 0.5 mL microtubes
4. Add 5 µL of DNA to the 0.5 mL microtube [from 20260620 Miniprep]
    a. Add 5 µL of DNA to the tube and label it respectively to its STK name (‘pSTK…”)
5. Incubate the labeled 0.5 mL microtubes in the MiniAmp thermocycler at 37C indefinitely until the laboratory is ready to experiment a gel electrophoresis to size the STK DNA

Done by Phoebe and Ethan Chang
Supervised by Gabriel
ELN by Ethan Chang

Date: 20260621
Title: Miniprep
Purpose: Prepare for electrophoresis
Procedure:
B3: pSTK-1A-5'-lacA
B4: pSTK-1D-3'-lacA
B5: pSTK-1A-5'-SacA
B6: pSTK-1D-3'-SacA
B7: pSTK-1A-5'-xvnP
B8: pSTK-1D-3'-xvnP
B9: pSTK-1A-Spacer
B10: pSTK-1B-Spacer
B11: pSTK-1C-Spacer
B12: pSTK-1D-Spacer

1. Transfer 1.8 mL of TB Medium to 2 mL microtube.
2. Centrifuge for 1 min at 16,000 g.
3. Discard the supernatant.
4. Repeat steps 1-3 until collected all medium.
   *Make sure the tip inside the TB medium tube is cleared.
5. Add 200 µL MX1 buffer.
6. Resuspend the pellet through vortexing or pipetting; cells should be completely resuspended.
7. Add 250 µL of MX2 buffer.
8. Gently mix by inverting the microtube 8-10 times.
9. Incubate the mixture at room temperature for 2 minutes.
10. Add 350 µL MX3 buffer.
11. Immediately and gently mix by inverting 4-5 times.
12. Centrifuge at 16,000 g for 5 minutes; meanwhile, place a Mini column onto a collection tube.
13. Transfer the supernatant carefully into the column.
14. Centrifuge at 7,000 g for 30 seconds. Discard the flow-through.
15. Wash the column once with 0.5 mL of WN Buffer by centrifuging at 7,000 g for 30 seconds. Discard the flow-through.
16. Wash the column once with 0.7 mL of WS buffer by centrifuging at 7,000 g for 30 seconds. Discard the flow-through.
17. Centrifuge the column at 10,000 g for another 3 minutes.
18. Place the column into a new 1.5 mL centrifuge tube.
19. Add 50 µL of elusion buffer onto the center of the membrane.
20. Stand the column for 2-3 minutes
21. Centrifuge at 10,000 g for 2-3 minutes to elute DNA.

Results:

	Concentration (ng/μL)	A260/A280
B3	134	1.836
B4	134	2.000
B5	139	1.904
B6	160	1.818
B7	160	1.951
B8	154	1.974
B9	148	1.827
B10	168	1.931
B11	178	1.894
B12	160	1.905

Done by Abby T and Abigail.
Supervised by Chars.
ELN by Sophia Y.

Date: 20260625
Title: 1.0% Agarose Gel
Purpose: Prepare 1.0% agarose gel for electrophoresis.
Procedure:

1. Assemble the mold and tray; prepare the comb.
2. Weight 0.2 g of agarose powder and obtain 20 mL 1X TAE solution with a graduated cylinder.
3. Place the contents into a conical flask.
4. Cover the flask with a plastic wrap then use a tip to puncture a few holes in the plastic wrap.
5. Microwave for 30 sec, then take it out and agitate it.
6. Microwave for 10 sec, then take it out and agitate it. Repeat until there are no particles in solution.
7. Wait for the gel to not smoke, then add 1 μL of safety dye and agitate it.
8. Pour the gel into the tray, then assemble the comb.
9. Wait for the gel to solidify.

Done by Sophia Y.
Recorded by Sophia Y.
Supervised by Chars.
ELN by Sophia Y.

Date: 20260625
Title: Golden Gate Assembly
Purpose: Assemble parts
Procedure:

1. Prepare the reaction as follow

A: (STK) Module 1: Pveg-RBS0-Ho1::pcyA-B0015

Vector 1A	0.176348 μL
Insert promoter (Pveg)	0.109618 μL
Insert RBS (RBS0)	0.1461117 μL
Insert CDS (Ho1::PcyA)	0.09070 μL
Insert terminator (B0015)	0.268869 μL
T4 DNA ligase buffer	2 μL
Sterile water	15.208353 μL
T4 DNA ligase	1 μL
BsaI-HFV2	1 μL
Total	20 μL

B: (STK) Module 2-1 (PliaG-RBS0-CcaSm3-B0015)

Vector 1B	0.1343046 μL
Insert promoter (PliaG)	0.2205 μL
Insert RBS (RBS0)	0.1461117μL
Insert CDS (CcaSm3)	0.071598 μL
Insert terminator (B0015)	0.268869 μL
T4 DNA ligase buffer	2 μL
Sterile water	15.1586167 μL
T4 DNA ligase	1 μL
BsaI-HFV2	1 μL
Total	20 μL

C: Module(2-2): Plep-RBS0-CcaSm3-B0015

Vector 1B	0.1343046 μL
Insert promoter (PlepA)	0.18795 μL
Insert RBS (RBS0)	0.1461117 μL
Insert CDS (CcaSm3)	0.071598 μL
`Insert terminator (B0015)	0.268869 μL
T4 DNA ligase buffer	2 μL
Sterile water	15.1911667 μL
T4 DNA ligase	1 μL
BsaI-HFV2	1 μL
Total	20 μL

D: Module(3): Pveg-RBS0-CcaR-B0015

Vector 1C	0.1428 μL
Insert promoter (Pveg)	0.109618 μL
Insert RBS (RBS0)	0.1461117 μL
Insert CDS (CcaR)	0.0392 μL
Insert terminator (B0015)	0.268869 μL
T4 DNA ligase buffer	2 μL
Sterile water	15.2934013 μL
T4 DNA ligase	1 μL
BsaI-HFV2	1 μL
Total	20 μL

E: Module(4): PcpcG2-veg-EEBD-MF001-csn:LEA-B0015

Vector 1D	0.153636 μL
Insert promoter (PcpcG2-veg)	0.047 μL
Insert RBS (EEBD-MF001)	0.1461117 μL

Insert CDS (csn::LEA)	0.03888 μL
Insert terminator (B0015)	0.268869 μL
T4 DNA ligase buffer	2 μL
Sterile water	15.3455033 μL
T4 DNA ligase	1 μL
BsaI-HFV2	1 μL
Total	20 μL

2. Use the PCR machine to run the program below:

temperature	time	cycle
37℃	1.5min	25-45cycle
16℃	3min	
37℃	5min	1cycle
60℃	10min	1cycle

Done by Alex

Date: 20260622
Title: Golden Gate Assembly
Purpose:Assemble parts
Procedure:

1. Preparing follow reaction:

A: (STK)Module 1:Pveg-RBS0-H01::PcyA-B0015

Vector 1A STK002-pSTK-1A-sfGFP	1.8 μL(dilute 1/10)
Insert promoter (Pveg)	1.1 μL(dilute 1/10)
insert+RBS (RBS0)	1.5 μL(dilute 1/10)
insert+CDS (Ho1::PcyA)	1.8 μL (dilute 1/20)
Insert+terminator (B0015)	1.3 μL (dilute(⅕)
T4 DNA ligase buffer	2 μL
Sterile water	8.5 μL
T4 DNA ligase	1 μL
BsaI-HFV2	1 μL
Total volume	20 μl

B:(STK) Module 2-1 PliaG-RBS0-CcaSm3-B0015

Vector 1B (sTK002-pSTK-1B-sfGFP)	1.3 μL(dilute 1/10)
Insert promoter (PliaG)	1.1 μL(dilute 1/5)
insert+RBS (RBS0)	1.5 μL(dilute 1/10)
insert+CDS (CcaSm3)	1.4 μL (dilute 1/20)
Insert+terminator (B0015)	1.4 μL (dilute(1/5)
T4 DNA ligase buffer	2 μL
Sterile water	9.3 μL
T4 DNA ligase	1 μL
BsaI-HFV2	1 μL
Total volume	20 μl

C:(STK)Module 2-2 PlepA-RBS0-CcaSm3-B0015

Vector 1B	1.3 μL(dilute 1/10)
Insert promoter (PlepA)	1.9 μL(dilute 1/5)
insert+RBS (RBS0)	1.5 μL(dilute 1/10)
insert+CDS (CcaSm3)	1.4 μL (dilute 1/20)
Insert+terminator (B0015)	2.7 μL (dilute(1/5)
T4 DNA ligase buffer	2 μL
Sterile water	7.2 μL
T4 DNA ligase	1 μL
BsaI-HFV2	1 μL
Total volume	20 μl

D:(STK)Module 3-1 Pveg-RBS0-CcaR-B0015

Vector 1C	1.4 μL(dilute 1/10)
Insert promoter (Pveg)	1.1 μL(dilute 1/5)
insert+RBS (RBS0)	1.5 μL(dilute 1/10)
insert+CDS (CcaR)	1.2 μL (dilute 1/20)
Insert+terminator (B0015)	1.3 μL (dilute(⅛)
T4 DNA ligase buffer	2 μL
Sterile water	9.5 μL
T4 DNA ligase	1 μL
BsaI-HFV2	1 μL
Total volume	20 μl

E: Module 4-1 Pveg-EEBD-MF001-csn::LEA-B0015

Vector 1D (sTK002-pSTK-1B-sfGFP)	2.2 μL(dilute 1/10)
Insert promoter (EEBD-MF002)	1.7 μL(dilute 1/5)
insert+RBS (csn::LEA)	2.5 μL(dilute 1/10)
insert+CDS (CcaSm3)	1.9 μL (dilute 1/20)

Insert+terminator (B0015)	2.5 μL (dilute/5)
T4 DNA ligase buffer	2 μL
Sterile water	5.3 μL
T4 DNA ligase	1 μL
BsaI-HFV2	1 μL
Total volume	20 μl

F: Module 4-2 Pveg-EEBD-MF001-csn::AmilCP-B0015

Vector 1D	2.2 μL(dilute 1/10)
Insert promoter (Pveg)	1.7 μL(dilute 1/5)
insert+RBS (EEBD-MF001)	2.5 μL(dilute 1/10)
insert+CDS (csn::AmilCP)	1.3 μL (dilute 1/20)
Insert+terminator (B0015)	2.5 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL
Sterile water	5.9 μL
T4 DNA ligase	1 μL
BsaI-HFV2	1 μL
Total volume	20 μl

G:(STK) module 4-1 PcpcG2-veg-EEBD-RBS0-csn::LEA-B0015

Vector 1D	1.5 μL(dilute 1/10)
Insert promoter (PcpcG2-veg)	1.4 μL(dilute 1/5)
insert+RBS (EEBD-RBS0)	2.1 μL(dilute 1/10)
insert+CDS (csn::LEA)	1.6 μL (dilute 1/20)
Insert+terminator (B0015)	1.3 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL
Sterile water	8.1 μL
T4 DNA ligase	1 μL

BsaI-HFV2	1 µL
Total volume	20 µl

H:(STK)module 4-2 (PcpcG2-veg-EEBD-RBS0-csn:ACCD-B0015

Vector 1D	1.5 µL (dilute 1/10)
Insert promoter (PcpcG2-veg)	1.4 µL (dilute 1/5)
insert+RBS (EEBD-RBS0)	1.0 µL (dilute 1/10)
insert+CDS (csn::AccD)	1.2 µL (dilute 1/20)
Insert+terminator (B0015)	1.3 µL (dilute(⅕)
T4 DNA ligase buffer	2 µL
Sterile water	9.4 µL
T4 DNA ligase	1 µL
BsaI-HFV2	1 µL
Total volume	20 µl

I:(STK) module 4-3 pcpcG2-veg-EEBD-RBS0-csn:AmilCP-B0015

Vector 1D	1.5 µL (dilute 1/10)
Insert promoter (PcpcG2-veg)	1.4 µL (dilute 1/5)
insert+RBS (EEBD-RBS0)	1 µL (dilute 1/10)
insert+CDS (csn:AmilCP)	1.4 µL (dilute 1/20)
Insert+terminator (B0015)	1.3 µL (dilute 1/5()
T4 DNA ligase buffer	2 µL
Sterile water	9.2 µL
T4 DNA ligase	1 µL
BsaI-HFV2	1 µL
Total volume	20 µl

Temperature	Time	Cycles

37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

Done by: Chloe, Abby T, Abigail, Abby K
Supervised by: Gabriel
ELN: Abby T, Sophie L

Date: 20260622
Title: Colony PCR of 06/17 (transformation pTWIST-Amp-High SamyQ-TAT-HA-AmilCP)
Purpose: Amplify arrived DNA parts.
Procedure:

1. Prepare the master mix x5 (5 reactions) 1-5 as follows:

	1 reaction	5 reactions
Forward Primer	1 µL	5 µL
Reverse Primer	1 µL	5 µL
Q-Amp Master Mix	10 µL	50 µL
ddH_2O	8 µL	40 µL

   FW primer: bio brick prefix
   Reverse primer: biobrick suffix
2. Dispense Solution (reaction) to 5 PCR tubes
3. Pick up a total of 5 colonies (each colony for each rxn) with the tip (circle the chosen colonies on the Petri dish before picking)
4. Releases the tips with the colonies in the PCR tubes and wait for 5 mins
5. Take out the tips and put tubes in PCR machines thermal cycle for 35 cycles:

95°C	5 min
95°C	30 sec
55°C	30 sec
72°C	1 min
72°C	5 min

Done by: Abigail, Abby K, Abby T
Supervised by: Gabriel

ELN: Sophie L

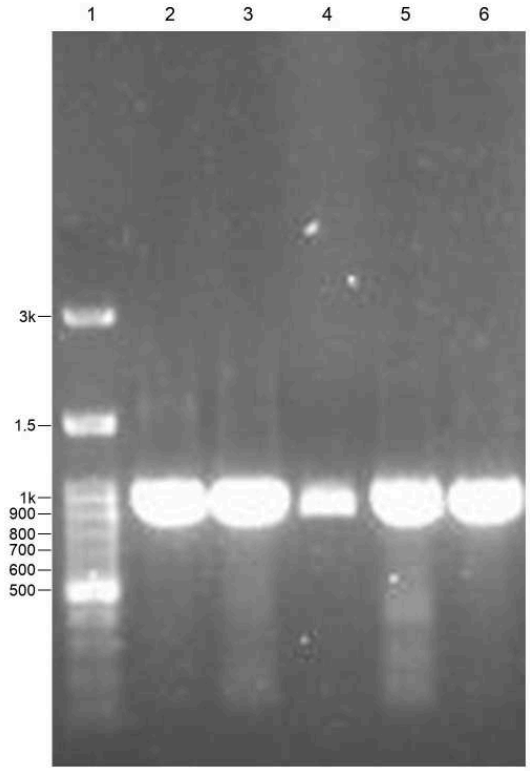
Date: 20260622
Title: Make 1% Agarose Gel
Purpose: Make 1% agarose gel for electrophoresis
Procedure:

1. Prepare 0.40 g agarose powder and 40 mL 1X TAE.
2. Pour it into a flask.
3. Use plastic wrap to cover the flask.
4. Use a pipette tip to poke a few holes in the plastic wrap.
5. Microwave for 30 seconds and agitate.
6. Microwave for 10 seconds and agitate; repeat this step until there are no sediments left in the solution.
7. Prepare the mold, tray, and comb while waiting for the solution to stop smoking.
8. Add 2 µL of safety dye and agitate.
9. Pour gel into the tray.
10. Wait until the gel solidifies.

Done by: Abigail
ELN: Sophie L

Date:20260623
Title: Gel Electrophoresis of 20260622 Colony PCR Results
Purpose: check the 20260622 colony PCR results
Procedure:

1. Load the gel in the layout shown below (5 μL each):
2. Resolve at 100 v for 30 min



Lane	Sample	Volume(μL)
1	[DM2300] ExcelBand™ 100 bp+3K DNA Ladder	5
2	pTWIST-Amp-High SamyQ-TAT-HA Amil CP	5
3	pTWIST-Amp-High SamyQ-TAT-HA Amil CP	5
4	pTWIST-Amp-High SamyQ-TAT-HA Amil CP	5
5	pTWIST-Amp-High SamyQ-TAT-HA Amil CP	5

6	pTWIST-Amp-High SamyQ-TAT-HA Amil CP	5

Done by: Abby K
Supervised by: Gabriel
ELN: Sophie L

Date:20260623
Title: Miniprep
Purpose: to amplify CCP Amil
Procedure:

1. Transfer 1.8 mL of medium to a 2 mL microtube
2. Centrifuge at 16000 g for 1 min
3. Discard the supernatant
4. Repeat steps 1-3 until collected all medium
5. Add 200 µL of MX1 buffer
6. Resuspend the pellet through vortexing or pipetting
7. The cells should be completely resuspended
8. Add 250 µL of MX2 buffer
9. Gently mix the buffer well by inverting 8-10 times
10. Incubate the mixture at room temperature for 2 min
11. Add 350 µL MX3 buffer
12. Immediately and gently mix by inverting 4-5 times
13. Centrifuge at 23444 g for 8 min
14. While centrifuging, assemble the column and collection tube
15. Transfer the supernatant(800µL) to the column using 200µL micropipette
16. Centrifuge at 7000 g for 30 sec
17. Discard the flowthrough
18. Add 500 µL WN buffer into the column
19. Centrifuge at 7000 g for 30sec
20. Discard the flowthrough
21. Add 700 µL WS buffer
22. Centrifuge at 7000 g for 30sec
23. Discard the flow-through
24. Centrifuge the empty column at 10000 g for 9 min to remove residual ethanol
25. Place the column into a new 1.5 mL microtube
26. Add 50µL sterile distilled water in the center of the membrane
27. Incubate the column at room temperature for 3 min
28. Centrifuge at 10000 g for 3 min

	Concentration (ng/µL)	A260/230	A260/180
A	70	0.805	1.795

B	101	0.944	1.870
C	134	0.905	1.861
D	104	0.819	1.825
E	103	0.990	1.907

Done by Chloe, Abby K
Supervised by Gabriel

Date:20260623
Title:Colony PCR
Purpose: Amplify DNA
Procedure:

1. Prepare the master mix x10

	1 rxn	10 rxn
Forward primer (Biobrick prefix)	1	10
Reverse primer (CDS+reverse)	1	10
Q-Amp master mix	10	100
ddH_2O	8	80

Plate A

	2 rxn
Forward (biobrick prefix)	2 µL
Reverse (pcyA)	2 µL
A-amp master mix	20 µL
ddH_2O	16 µL

Plate B

	2 rxn
Forward	2 µL

(biobrick prefix)	
Reverse (CcASm3)	2 µL
A-amp master mix	20 µL
ddH_2O	16 µL

Plate C

	2 rxn
Forward (biobrick prefix)	2 µL
Reverse (CcaSm3)	2 µL
A-amp master mix	20 µL
ddH_2O	16 µL

Plate G

	2 rxn
Forward (biobrick prefix)	2 µL
Reverse (LEA)	2 µL
A-amp master mix	20 µL
ddH_2O	16 µL

Plate H

	1 rxn
Forward (biobrick prefix)	1 µL
Reverse (AccD)	1 µL
A-amp master mix	10 µL
ddH_2O	8 µL

Plate I

	1 rxn
Forward (biobrick prefix)	1 µL
Reverse (pcyA)	1 µL
A-amp master mix	10 µL
ddH_2O	8 µL

2. Dispense solution (reaction into 10 PCR tubes
3. Pick up a total of 10 colonies with the tip
4. Release the tips with the colonies in the PCR tubes+wait 5 min
5. Place tubes in PCR machines

Done by Abby K, Chloe

Date:20260623
Title:prepare 1.0% agarose gel for electrophoresis
Purpose: prepare gel for electrophoresis
Procedure:

1. Assemble the mole & tray
2. Weigh 0.2 g of agarose powder and add it into a flask
3. Add 20mL of 1XTAE solution into the flask
4. Cover the flask with plastic wrap and punctuate some holes
5. Microwave for ~3-seconds and agitate it. Repeat this step until all particles are dissolved
6. Wait until the gel stops smoking but is still hot
7. Add 1 µL of safety dye into the gel and agitate it
8. Pour the gel into the tray, then assemble the comb
9. Wait until gel dried

Done by Jacquelyn
Supervised by Gabriel

Date:20260623
Title:Golden Gate assembly
Purpose: To assemble the light sensor module
Procedure:

1. Prepare the reaction using following protocol

V:vecter
P:powder
R:RBS
C:cds

T:terminater

A:Pveg-RBSO-Ho1::pcyA-B0015 (1A) (STK)

V	115 μL	3120 bp
P	280 μL	3361 bp
R	188 μL	3113 bp
C	222 μL	1549 bp
T	107 μL	2214 bp
total		11356 bp

B: PliaG-RBS0-CcaSm3-B0015 (1B) (STK)

V	151 μL	3120 bp
P	130 μL	2205 bp
R	188 μL	2113 bp
C	266 μL	1465 bp
T	107 μL	2213 bp
total		11066 bp

C: PlepA-RBS0-CcaSm3-B0015 (1B) (STK)

V	151 μL	3120 bp
P	153 μL	2241 bp
R	188 μL	3113 bp
C	266 μL	1465 bp
T	107 μL	2213 bp
total		11152 bp

D:Pveg-RBS0-CcaR-B0015 (1C) (STK)

V	142 μL	3120 bp
P	280 μL	2361 bp
R	188 μL	2113 bp
C	253 μL	760 bp

T	107 µL	2213 bp
total		10567 bp

E:Pveg-EEBD-MF001-csn::LEA-B0015 (1D) (VoClo2.0)

V	114 µL	3815 bp
P	183 µL	271 bp
R	168 µL	3224 bp
C	195 µL	2797 bp
T	66 µL	2520 bp
total		12627 bp

F:Pveg-EEBD-MF001-csn::AmilCP-B0015 (1D)

V	114 µL	3815 bp
P	183 µL	271 bp
R	168 µL	3224 bp
C	305 µL	2986 bp
T	66 µL	2520 bp
total		12816 bp

G:PcpcG2-veg-EEBD-csn::LEA-B0015 (1D) (STK)

V	132 µL	3120 bp
P	65 µL	235 bp
R	150 µL	2392 bp
C	212 µL	634 bp
T	107 µL	2213 bp
total		8594 bp

H:PcpcG2-veg-EEBD-riboj-RBS0-csn::ACCD-B0015 (1D) (STK)

V	132 µL	3120 bp
P	65 µL	235 bp

R	150 µL	2392 bp
C	252 µL	1195 bp
T	107 µL	2213 bp
total		9155 bp

I: PcpcG2-veg-EEBD-riboj-RBS0-csn::AmilCP-B0015 (1D) (STK)

V	132 µL	3120 bp
P	65 µL	235 bp
R	150 µL	2342 bp
C	150 µL	823 bp
T	107 µL	2213 bp
total		8783 bp

J:Pveg-EEBD-riboj-RBS0

V	132 µL	3120 bp
P	280 µL	2362 bp
R	150 µL	2392 bp
C	250 µL	823 bp
T	107 µL	2213 bp
total		10904 bp

2. For STK using Amp and for MoClo 2.0 using Kana
3. Assemble protocol

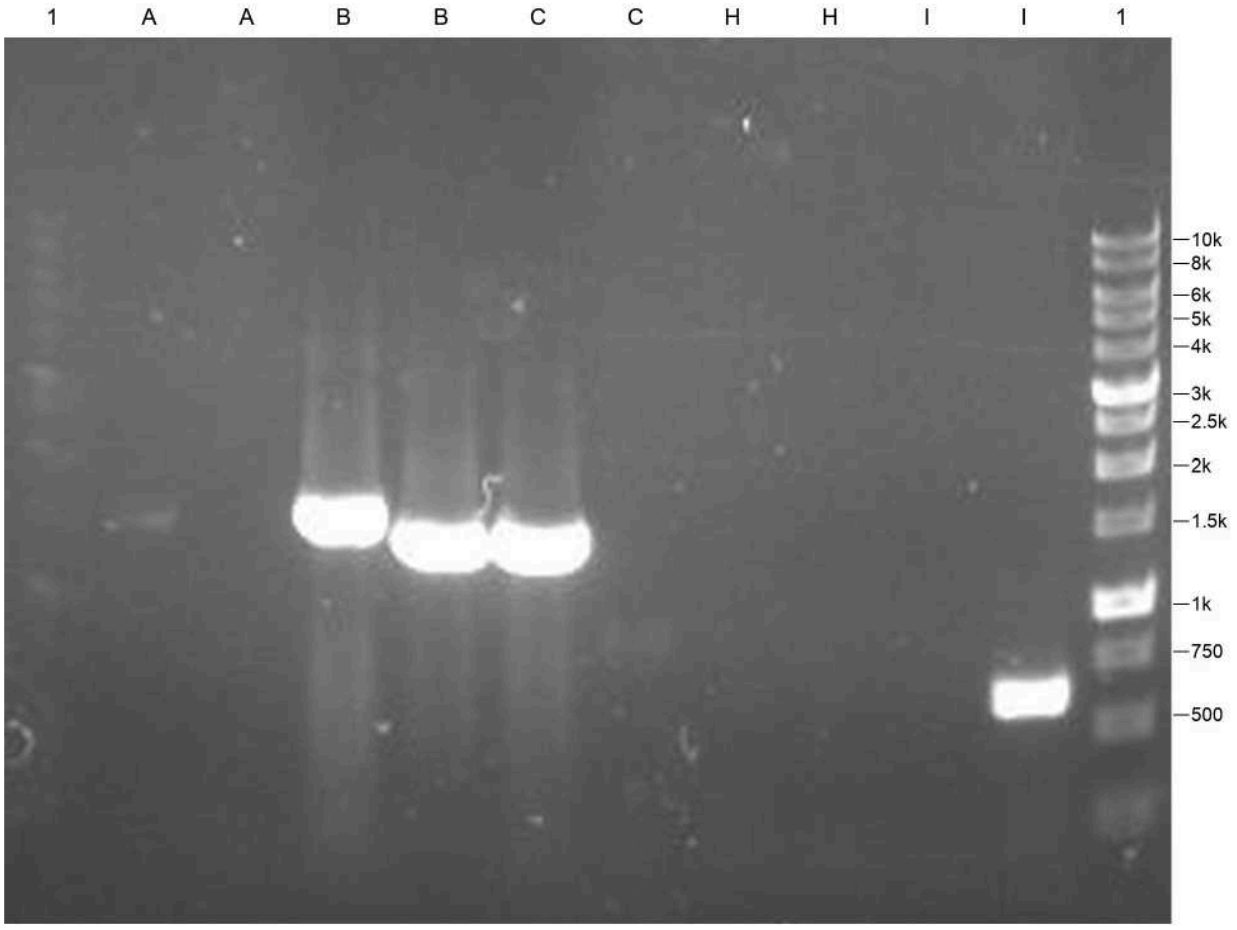
37 ℃	1 min	30 cycles
16 ℃	1 min	
60 ℃	5 min	1 cycle

4. Transformation protocol follow 2026/06/02

Done by Gabriel

Date: 20260624
Title: Electrophoresis
Purpose:check PCR result
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



1 = Omics 1kb DNA RTU Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A: Pveg - RBS0 - Ho1::PcyA - B0015	Biobrick F	CcaSm3 R	1602
B: PliaG - RBS0 - CcaSm3 - B0015	Biobrick F	CcaSm3 R	1602

C: PlepA - RBS0 - CcaSm3 - B0015	Biobrick F	CcaSm3 R	1602
H: PcpcG2veg - EEBD - RiboJ - RBS0 - csn::ACCD - B0015	Biobrick F	CcaSm3 R	1602
I: PcpcG2veg - EEBD - RiboJ - RBS0 - csn::AmilCP - B0015	Biobrick F	CcaSm3 R	1602
Note: B: Positive clone			

Done by Ethan C.
ELN by Sophia Y.

Date: 20260624
Title: Making 1% agarose gel
Purpose: Confirm colony PCR sizes.
Procedure:

1. Collect 0.4 g of agarose powder and 40 mL of 1X TAE.
2. Pour it into a flask.
3. Use a piece of plastic wrap to cover the flask.
4. Use a pipette tip to poke a few holes in the plastic wrap.
5. Microwave for 30 seconds.
6. Take out to agitate.
7. Microwave for 5 seconds.
8. Take out to agitate.
9. Repeat 7-8 until no more sediments.
10. Wait for the mixture to be warm but not smoking.
11. Meanwhile, assemble the tray and comb.
12. Add 2 µL of safety dye.
13. Pour it into the tray.
14. Wait to solidify at room temperature.

Done by Abby K and Sophie L.
Supervised by Chars.
ELN by Sophia Y.

Date: 20260624
Title: TA cloning
Purpose: Insert PCR products (MoClo → STK) to pGEM-Teasy.
Procedure:

1. Prepare reactions by the following table:

Code	PCR products	Conc. (ng/μL)	Insert volume	2x buffer
A	Csn: Lea	21.2	0.6	5
B	Csn: AmilCP	15.0	1.2	5
C	Csn: ACCD	25.2	1.1	5
D	CcaSm3	26.6	1.3	5
E	CcaR	25.2	0.7	5
F	Ho1:pcyA	22.2	1.7	5
G	PcpcG2-Pveg	6.5	0.7	5

2. Incubated at 4°C since 23:00

Done by Chars.
ELN by Sophia Y.

Date: 20260625
Title: Colony PCR
Purpose: amplify DNA
Procedure:

1. Prepare the master mix (25 reactions).
2. Pipette 21 µL of the master mix into 8-strip-PCR-tubes.
3. Circle three colonies from the plates

A: 1A-Preg-RBS0-Ho1::PcyA-B0015
B: 1C-Preg-RBS0-ccaR-B0015
C: 1D-PcpcG2-veg-EEBD-RBS0-Csn::ACCD-B0015
D: 1D-PcpcG2-veg-EEBD-RBS0-Csn::LEA-B0015
E: 1B-PlepA-RBS0-ccasm3-B0015
F: 1B-PliaG-RBS0-ccasm3-B0015
G: 1D-PcpcG2-veg-EEBD-RBS0-Csn::AmilCP-B0015
H: 1D-Pveg-EEBD-RBS0-Csn::AmilCP-B0015

4. Mark the agar plate (ampicillin) by dividing it into 3 rows of 8.
5. For each colony, pick it up with a pipette tip and swipe it on its respective spot on the agar plate, then release the tip into its respective microtube in the 8-strip-PCR-tube.
6. Take out the tips and close the lid
7. Place into PCR machine for thermal cycle

95°C	2 min	1 cycle
95°C	30 sec	30 cycles
55°C	20 sec	
72°C	1 min	
72°C	5 min	1 cycle

Done by Sophie L, Sophia Y
Recorded by Sophia Y
Supervised by Venus

Date: 20260625
Title: Making 1% Agarose Gel
Purpose: Make 1% agarose gel to confirm colony PCR sizes using electrophoresis.
Procedure:

1. Prepare 0.40 g agarose powder and 40 mL 1X TAE.
2. Pour it into a flask.

3. Use plastic wrap to cover the flask.
4. Use a pipette tip to poke a few holes in the plastic wrap.
5. Microwave for 30 seconds and agitate.
6. Microwave for 10 seconds and agitate; repeat this step until there are no sediments left in the solution.
7. Prepare the mold, tray, and comb while waiting for the solution to stop smoking.
8. Add 2 μL of safety dye and agitate.
9. Pour gel into the tray.
10. Wait until the gel solidifies.

Done by Sophie L and Sophia Y.
Recorded by Sophia Y.
ELN by Sophia Y.

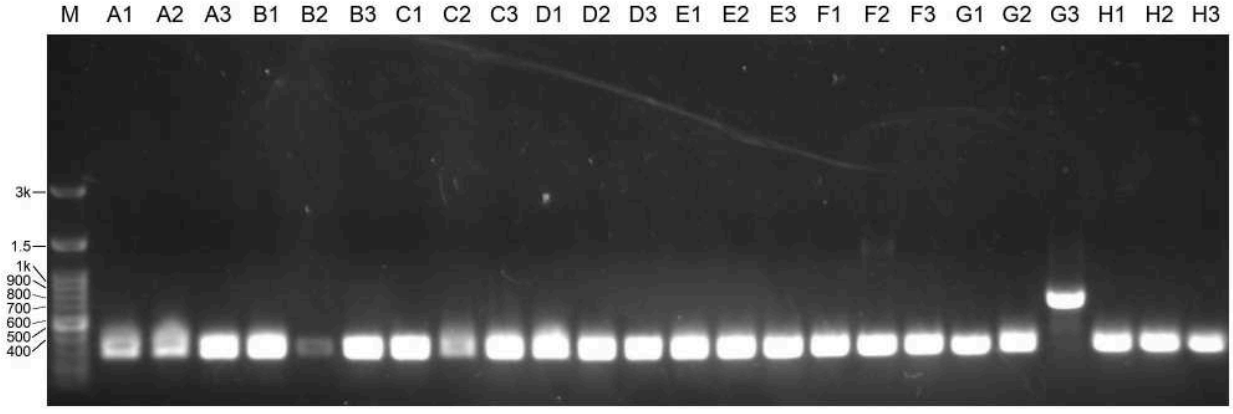
Date: 20260625
Title: SOC Medium Preparation
Purpose: prepare SOC medium
Procedure:

1. Prepare 3.17 g of SOC broth powder.
2. Prepare 100 mL ddH_2O
3. Mix together by using a stirring rod.

Done by Chars.
Recorded by Chloe.
ELN by Sophia Y.

Date: 20260625
Title: Gel electrophoresis
Purpose: Checking Colony PCR result
Procedure:

1. Load the gel in the layout shown below (5 μL each):
2. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
Plasmid Construct
A: pSTK-1A-Pveg-RBS0-Ho1:pcyA-B0015
B: pSTK-1C-Pveg-RBS0-CcaR-B0015
C: pSTK-1D-PcpcG2-veg-EEBD-Csn:Accd-B0015
D: pSTK-1D-PcpcG2-veg-EEBD-Csn:Lea-B0015
E: pSTK-1B-PlepA-RBS0-CcaSm3-B0015
F: pSTK-1B-PliaG-RBS0-CcaSm3-B0015
G: pSTK-1D-PcpcG2-veg-EEBD-Csn:AmilCP-B0015
H: pSTK-1D-Pveg-EEBD-Csn:AmilCP-B0015

Done by Abby K.
Supervised by Chars.
ELN by Sophia Y.

Date: 20260625
Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:

1. Turn on the water bath, set it to 42 °C and prepare ice
2. Mix 100 μL competent cell with 10 μL plasmid

A: pSTK-1A-Pveg-RBS0-Ho1:pcyA-B0015
B: pSTK-1C-Pveg-RBS0-CcaR-B0015
C: pSTK-1D-PcpcG2-veg-EEBD-Csn:Accd-B0015
D: pSTK-1D-PcpcG2-veg-EEBD-Csn:Lea-B0015
E: pSTK-1B-PlepA-RBS0-CcaSm3-B0015
F: pSTK-1B-PliaG-RBS0-CcaSm3-B0015
G: pSTK-1D-PcpcG2-veg-EEBD-Csn:AmilCP-B0015

3. Incubate on ice for 20min
4. Prepare reaction for blue-white screening
   A . 1000 μL of ddH_2O + 4 μL of X-Gal (20 ng/mL) (final concentration 80 mg/mL)
   B . Spread on LBA plate
5. Heat shock at 42 °C for 45 sec
6. Incubate on ice for 3 min
7. Add 1mL of SOC medium and incubate at 37 °C for 1 hour

8. Centrifuge at 5000 g for 5min and remove 1 ml of the supernatant
9. Resuspend the cell pellet in the remaining supernatant and spread the mixture onto the LBA plate
10. Place the plates into the incubator after labeling

Done by Chloe, Abby K
Supervised by Chars
Recorded by Sophie H.

Date: 20260626
Title: Electrophoresis Gel Preparation
Purpose: Prepare gel for electrophoresis
Procedure:

1. Prepare and assemble the mold and comb
2. Weight 0.4 g of agarose powder and add 40 ml of 1X TAE solution into flask
3. Cover the flask with plastic wrap, then use a tip to puncture holes on the plastic wrap
4. Microwave until the gel stops smoking but is still hot
5. Add 2 µL of safety dye into the gel and agitate it
6. Pour the melted gel into the tray
7. Wait until the gel cools down

Done by Abby K Sophic L
Supervised by Chars

Date: 20260626
Title: Colony PCR
Purpose:amplify DNA
Procedure:

1. Prepare the following reaction (master mix) X100 (100 reactions) and dispense solution into PCR tubes.

	1 reaction	100 reactions
Forward Primer (pSTKseq-fw 10 mM)	1 µL	100 µL
Reverse Primer (pSTKseq-rv 10 mM)	1 µL	100 µL
Qamp Master Mix	10 µL	1000 µL
ddH_2O	8 µL	800 µL

2. Circle 12 colonies form each of the plates.

A: 1A-Pveg-RBS0-Ho1::PcyA-B0015
B: 1C-Pveg-RBS0-CcaR-B0015
C: 1D-PcpcG2veg-EEBD-RBS0-csn::ACCD-B0015
D: 1D-PcpcG2veg-EEBD-RBS0-csn:LEA-B0015
E: 1B-PlepA-RBS0-Ccasm3-B0015
F: 1B-PliaG-RBS0-Ccasm3-B0015
G: 1D-PcpcG2veg-EEBD-RBS0-csn:AmilCP-B0015
H: 1D-Pveg-EEBD-RBS0-csn:AmilCP-B0015

3. Pick up 12 colonies per plate with a tip and streak the labelled grid.
4. Dispense the tips with colonies into the PCR strip and wait 5 min.

5. Take out the tips and close the lid.
6. Place into PCR machine for the thermal cycle.

Done by Abigail, Abby T, Abby K, Chloe, Sophie L.
Supervised by Chars.
ELN by Sophia Y.

Date: 20260626
Title: Making 1% Agarose Gel
Purpose:prepare agarose gel for electrophoresis
Procedure:

1. Prepare a 0.20 g agarose powder and 20 mL 1X TAE.
2. Pour it into a flask.
3. Use plastic wrap to cover the flask.
4. Use a pipette tip to poke a few holes in the plastic wrap.
5. Microwave for 30 sec and agitate.
6. Microwave for 10 sec and agitate until there are no sediments left.
7. Prepare the mold, tray, and comb while waiting for the solution to stop smoking.
8. Add 1 µL of safety dye and agitate.
9. Pour into the tray.
10. Wait until the gel solidifies.

Done by Sophie L.
Supervised by Gabriel.
ELN by Sophia Y.

Date: 20260626
Title: Minprep
Purpose: to isolate plasmid
Procedure:
A: pSTK-1B PliaG: Ccasm3 #1
B: pSTK-1B PliaG: Ccasm3 #2
C: pSTK-1B Plepa: Ccasm3 #1
D: pSTK-1A Pveg:Ho1:pcyA #3
E: pSTK-1D PcpcG2-veg:csn:AmilCP #2
F. pSTK-1C Pveg:csn:AmilCP #1

1. Transfer 1.8 mL of TB medium to 2 mL microtube.
2. Centrifuge for 1 min at 16,000 g.
3. Discard the supernatant.
4. Repeat steps 1-3 until collected all medium.
5. Add 200 µL of MX1 Buffer to the pellet and resuspend the cells completely by vortexting or pipetting.
6. Add 250 µL of MX2 Buffer.
7. Gently mix by inverting the tube 8-10 times.

8. Incubate at room temp for 2 minutes.
9. Add 350 µL of MX3 Buffer and immediately and gently mix the solution well.
10. Centrifuge at 10,000 g for 5 minutes. Meanwhile, place a MiniPlus column onto a collection tube.
11. Transfer the supernatant onto the column.
12. Centrifuge at 7,000 g for 30 seconds; discard the flow through.
13. Wash the column once with 500 µL of WN buffer by centrifuging at 7,000 g for 30 seconds; discard the flow through.
14. Wash the column once with 700 µL of WS buffer by centrifuging at 7,000 g for 30 seconds. Discard flow through.
15. Centrifuge column at 1,000 g for another 3 mins.
16. Place column into new 1.5 mL centrifuge tube.
17. Add 50 µL of Elution Buffer onto the center of the membrane.
18. Stand the column for 2-3 mins and centrifuge at 10,000 g for 3 mins.
19. Test concentration at Biodrop.

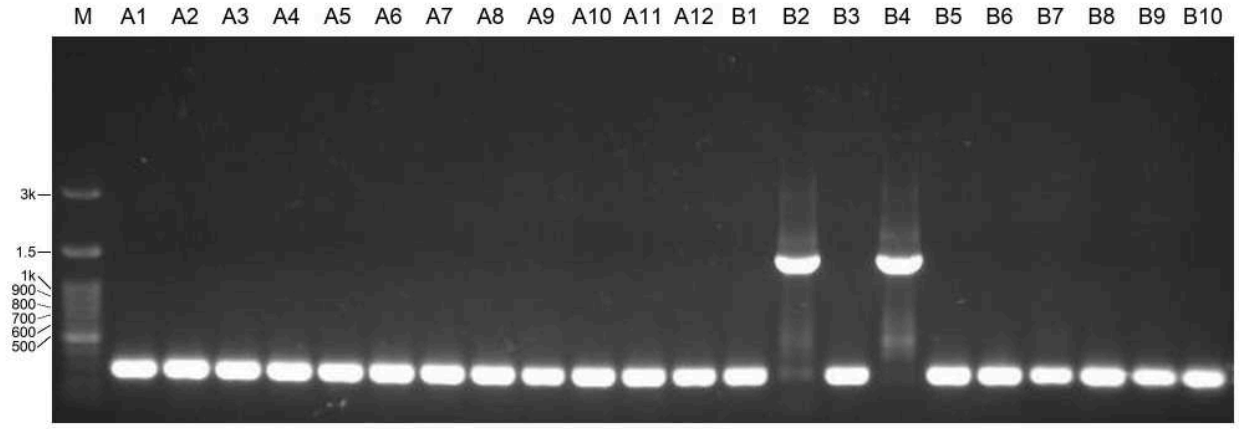
Results:

	Concentration (ng/µL)	A260/A280
A	78.00	2.053
B	86.00	1.870
C	80.00	2.000
D	98.00	1.922
E	111.0	1.790
F	64.00	1.882

Done by Abigail, Abby T, and Chloe.
Supervised by Chars.
ELN by Sophia Y.

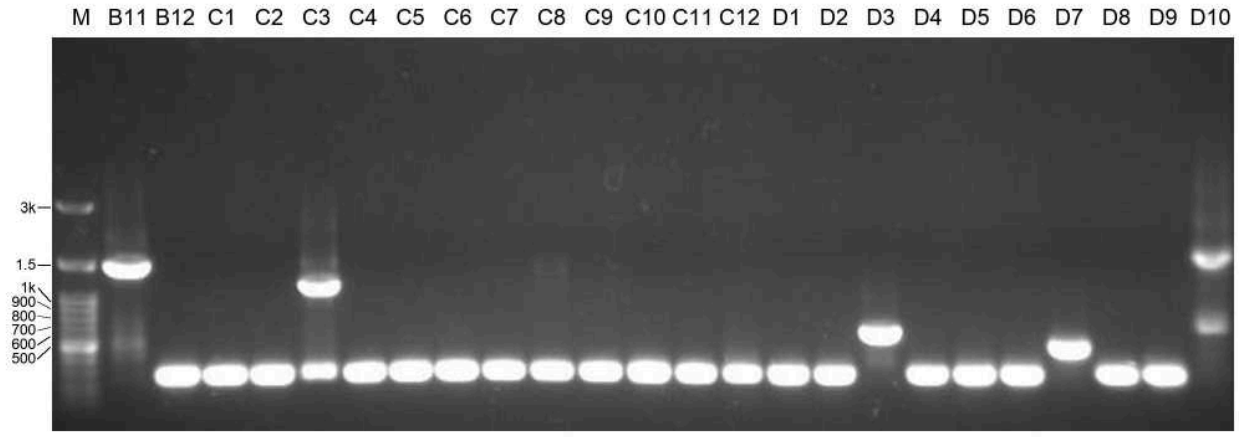
Date: 20260626
Title: Gel electrophoresis
Purpose:check PCR result
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
Plasmid Construct
A: 1A-Pveg-RBS0-H01::PcyA-B0015
B: 1C-Pveg-RBS0-CCaR-B0015
Note: B2, B4 Positive clone

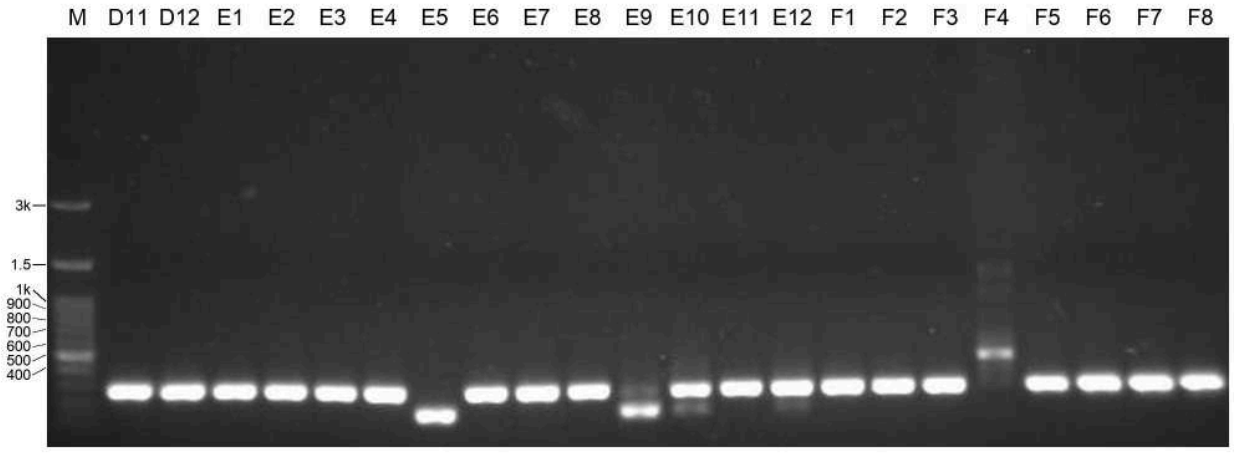
3. the gel in the layout shown below (5 μL each):
4. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
Plasmid Construct
B: 1C-Pveg-RBS0-CCaR-B0015

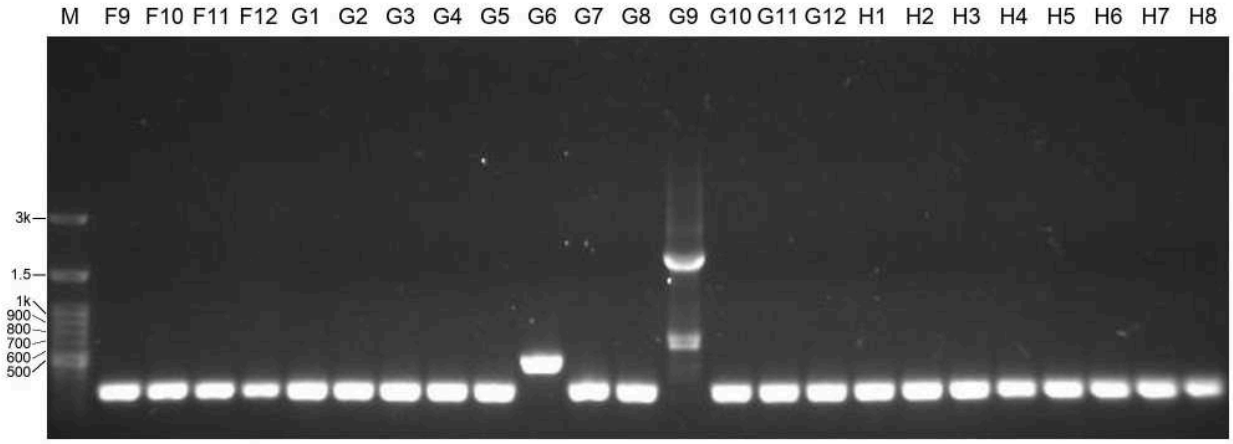
C: 1D-PcpcG2-veg-EEBD-RBS0-Csn::ACCD-B0015
D: 1D-PcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015

5. Load the gel in the layout shown below (5 µL each):
6. Resolve at 100 v for 30 min



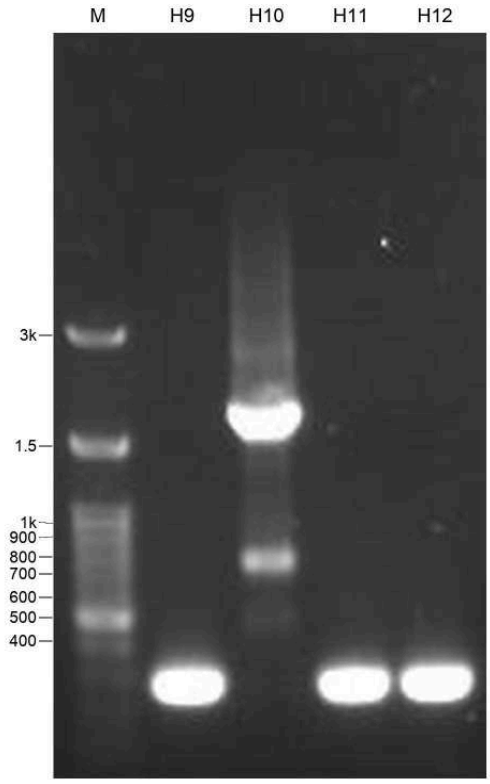
M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
Plasmid Construct
D: 1D-PcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015
E: 1B-PlepA-RBS0-Ccasm3-B0015
F: 1B-PliaG-RBS0-Ccasm3-B0015

7. Load the gel in the layout shown below (5 µL each):
8. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
Plasmid Construct
F: 1B-PliaG-RBS0-Ccasm3-B0015
G: 1D-PcpcG2veg-EEBD-RBS0-Csn:AmilCP-B0015
H: 1D-Pveg-EEBD-RBS0-Csn:AmilCP-B0015

9. Load the gel in the layout shown below (5 μL each):
10. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL
Plasmid Construct
H: 1D-Pveg-EEBD-RBS0-Csn:AmilCP-B0015
Note: H10 Positive clone

Done by Sophie L and Sarah C.
ELN by Sophia Y.

Date: 20260627
Title: Miniprep
Purpose: To isolate DNA for amplification
Procedure:

1. Transfer 1.8mL TB medium to a 2 mL microtube

B2 : 1C-Pveg-RBS0-CCaR-B0015 #2
B4 : 1C-Pveg-RBS0-CCaR-B0015 #4
C3 : 1D-PcpcG2veg-EEBD-RBS0-csn::ACCD-B0015 #3
D10: 1D-PcpcG2veg-EEBD-RBS0-csn::LEA-B0015 #10
G9 : 1D-PcpcG2veg-EEBD-RBS0-csn::AmilCP-B0015 #9

2. Centrifuge at 16000 g for 1 minute
3. Discard the supernatant
4. Repeat steps 2-3 until there is no TB medium left in the tube
5. Add 200 µL of MX1 buffer
6. Resuspend the pellet through vortexing or pipetting
7. The cell should be completely resuspended before moving to the next step
8. Add 250 µL of MX2 buffer
9. Gently mix the buffer well by inverting the microtube back and forth 8 times
10. Incubate the mixture at room temperature for 2 minutes
11. Add 350 µL of MX3 buffer to the mixture
12. Immediately and gently mix via inversion 4-5 times
13. Centrifuge the mixture at max ref (23.44 g) for 5 minutes
14. While centrifuge, assemble column and collection
15. Transfer supernatant 1800 µL to the column using a 200 µL micropipette to avoid transferring pellet and contaminating the supernatant
16. Centrifuge the column at 7000 g for 30 min
17. Discard the flow-through liquid in the collection tube
18. Add 500 µL of WN buffer
19. Centrifuge at 7000 g for 30 sec
20. Discard the flow through
21. Add 700 µl WS buffer
22. Centrifuge at 7,000 g for 30 seconds
23. Discard the flow-through
24. Centrifuge the empty column at 10,000 g for 3 minutes to remove residual ethanol
25. Place the column into a new 1.5 ml microtubule
26. Add 50 µl of EB buffer on the center of the membrane
27. Incubate the column at room temperature for 3 minutes
28. Centrifuge at 10,000 g for 3 minutes

Results:

	Concentration	A260/280	A260/230

	(ng/μL)		
B2	120.0	1.846	0.930
B4	69.0	1.971	1.353
D10	129.0	1.743	0.772
C3	90.0	1.837	1.011
G9	106.0	1.738	0.768

Done by Sarah , Ethan Chang , Naomi
Supervised by Chars

Date: 20260627
Title: PCR
Purpose: amplify the arrived DNA parts
Procedure:

1. DNA templates

Code	Name	Concentration (ng/μL)	Forward Primer	Reverse primer
A	ccaR	119	ccaRF	ccaR R
B	ccaSm3	187	Ccaw3	ccaSm3 R
C	csn:AmilCP	305	csnF	AmilCP R
D	csn::ACCD	278	csnF	ACCD R
E	csn::LEA	195	csnF	LEA R
F	Ho1-pcyA	211	H01F	Pcy A R
G	PcpcG2-veg	90	PcpcG2-vegF	PcpcG2-veg R
H	Cpp Amil 1	70	SamyQ	AmilCP R

2. Prepare the following

reagent	1 rxn (50 μL)	3 rxn (150 μL) x9 (master mix)
10x ExTaq buffer	5 μL	15 μL
2.5 mM dNTP mix	4 μL	12 μL

10 μM forward primer	1 μL	3 μL
DNA temp	1 μL	3 μL
10 μM reverse primer	1 μL	3 μL
PCR grade water	37.75 μL	113.25 μL
Ex Taq polymerase	0.25 μL	0.75 μL

3. Distribute 50μL of 3rxn mixture into each tube for the corresponding part
4. Place into PCR machine, applying the thermal cycle below:

98 ℃	30 sec
98 ℃	10 sec
60 ℃	30 sec
72 ℃	90 sec
72 ℃	5 min
23 ℃	0 min

Done by Audrey H, Abby T
Supervised by Chars

Date: 20260627
Title: Prepare 1% agarose gel
Purpose: extracting desired DNA gel electrophoresis
Procedure:

1. Assemble the mold and comb
2. Weight 0.2g of agarose powder and 20mL of 1XTAE into a flask
3. Cover the flask with a plastic wrap and poke 3 holes with a tip
4. Put the flask into the microwave for 30 sec
5. Take it out and agitate it
6. Repeat step 3 and 4 until all particles are dissolved
7. Take the flask out and cool down until its warm but still hot
8. Put 1 μL of safety dye into the flask
9. Pour the melted gel into the mold
10. Wait 20 min for the gel to solidify
11. Repeat steps 1-10 until 8 pieces of gel are made

Done by Audrey H, Abby T, Abby K , Sophie L
Supervised by : Chars

Date: 20260627
Title: Gel electrophoresis
Purpose: check the PCR result
Procedure:

1. Mix 50μL of PCR product from 1 tube with 10μL of loading dye
2. Load samples in the following layout

   Gel1
   M A M
   5 150 5
   Gel2
   M B M
   5 150 5
   Gel3
   M C M
   5 150 5
   Gel4
   M D M
   5 150 5
   Gel5
   M E M
   5 150 5
   Gel6
   M F M
   5 150 5
   Gel7
   M G M
   5 150 5
   Gel8
   M H M
   5 150 5
   1: 1kb DNA ladder
   A: CcaR
   B: CcaSm3
   C: can:AmilCP
   D: can:AccD
   E: can:LEA
   F: H01:pcya
   G: PcpcG2-veg
   H: CppAmil1
3. Resolve at 100 v for 30,in

Done by Abby K/T, Sophie L
Supervised by Chars

Date: 20260627
Title: gel extraction
Purpose: isolate desired gene fragments from PCR product
Procedure:

1. Under the UV light, identify if the DNA is the correct one
2. Use a knife to cut out the DNA bind
3. Cut up the vine into smaller pieces
4. Place the pieces into the 2mL microtube
5. Add 500 µL of GP buffer
6. Place the tube onto the thermal shaker at 800 rpm, 60 ℃

Done by Abby K/T, Sophie L
Supervised by Gabriel, Chars

Date: 20260628
Title: Miniprep
Purpose: isolate the plasmid
Procedure:
G4: pSTK-OC2-His6x
H7 : pSTK-OCx-Spacer
E8 : pSTK0-B-RBS2
H10 : (cPCR)

1. Transfer 1.8 mL of TB medium to 2 mL microtubes
2. Centrifuge for 1 min at 16000 g
3. Discard the supernatant
4. Repeat steps 1-3 until collected all medium
5. Add 200 µL of MX1 buffer
6. Resuspend the pellet through vortexing or pipetting
7. Add 250 µL of MX2 buffer
8. Gently mix by inverting the microtube 8-10 times
9. Incubate the mixture at room temperature of room temperature for 2 minutes
10. Add 350 µL MX3 buffer
11. Immediately and gently mix through inverting 4-5 times
12. Centrifuge at 16000 g for 5 minutes
13. Meanwhile, assemble the column and collection tube
14. Transfer the supernatant (around 800 µL) to the column using 200mL pipette
15. Centrifuge at 7000 g for 30 second, discard the flow through
16. Wash the column once with 500 µL of WN buffer by centrifuging at 7000 g for 30 seconds , discard the flow through
17. Wash the column with 700 µl of WS buffer. Centrifuge at 7000 g for 30 seconds. Discard the flow through
18. Centrifuge column for 8 mins at 10000 g
19. Place the column into new 1.5 mL centrifuge tubes
20. Add 50 µL of nuclear free water onto the center of membrane
21. Stand the column for 2-3 mins and centrifuge at 10000 g for 3 mins
22. Test concentration at biodrop

	Concentration(ng/µL)	A260/230	A260/280
G4	135.0	1.216	1.957
H7	127.0	0.722	1.868
E8	78.00	1.200	1.950
H10(cpCR)	98.06	0.907	1.885

Done by Abigail

Supervised by Chars

Date: 20260628
Title: Making liquid culture
Purpose: Amplify F2 and F8 from the subtiTookit
Procedure:

1. Label two liquid culture tubes with F2 and F8
2. Turn on the bunsen burner
3. Add 3 mL of LB medium into each tube
4. Add 5 µL of Chloramphenicol into each tube
5. Dip a medium pipette tip into the subtiToolKit respective agar
6. Dispense the tips(with agar) in their respective tubes
7. Place the tubes into the incubator at a tilted angle

Done by Abby K
Supervised by Gabriel , Chars

Date: 20260629
Title: Golden Gate Assembly
Purpose: To assemble multiple core genetic parts of the light-controlled optogenetic circuit from level 1 and level 2.
Procedure:

1. Insert: vector=2:1 pmols

A: (STK) Module 1(1A): Pveg–RBS0-Ho1:pcyA-B0015

1. prepare the reaction as follow:

Vector (1A)	1.8 μL (dilute 1/10)
Insert promoter (Pveg)	1.1 μL (dilute 1/10)
Insert RBS (RBS0)	1.5 μL (dilute 1/10)
Insert CDS (Ho1::PcyA)	1.7 μL (dilute 1/20)
Insert Terminator (B0015)	1.3 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL
Sterile water	8.6 μL
T4 DNA ligase	1 μL
BsaI-HFv2	1 μL
Total	20 μL

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

B: M4-Pveg - EEBD- RBS0 - CSN::ACCD - B0015

1. prepare the reaction as follow:

Vector (1D)	1.5 µL(dilute 1/10)
Insert promoter (Pveg)	1.1 µL (Dilute 1/10)
Insert RBS (EEBD-RBS0)	2.1 µL(dilute 1/10)
Insert CDS (CSN::ACCD)	1.5 µL(dilute 1/10)
Insert Terminator (B0015)	1.3 µL (dilute 1/5)
T4 DNA ligase buffer	2 µL
Sterile water	8.5 µL
T4 DNA ligase	1 µL
BsaI-HFv2	1 µL

Total	20 μL

2. Use the thermocycler to run the program below:

Temperature		Time	Cycles
37 °C (Restriction enzyme digestion)		1.5 minutes	25-45 cycles
16 °C (Ligation)		3 minutes	
37 °C (Restriction enzyme digestion)		5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)		10 minutes	1 cycle

C: M3-Pveg-RBS0-CcaR-B0015

1. prepare the reaction as follow:

Vector (1C)	1.4 μL(dilute 1/10)
Insert promoter(Pveg)	1.1 μL (dilute 1/10)
Insert RBS(RBS0)	1.5 μL (dilute 1/10)
Insert CDS(CcaR)	1.0 μL (dilute 1/10)
Insert Terminator (B0015)	1.3 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL
Sterile water	9.6 μL
T4 DNA ligase	1 μL

BsaI-HFv2	1 μL
Total	20 μL

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

D: Pveg-EEBD-RBS0-SamyQ::TAT-Ha::AmilCP-B0015

1. prepare the reaction as follow:

Vector (shuttle vector): pSTK-EXP-1-sfGFP (bp:6929)	1.4 μL (dilute ⅛)
Insert promoter(Pveg)	1.1 μL (dilute 1/10)
Insert RBS(EEBD-RBS0)	1.0 μL (Dilute 1/5)
Insert CDS(SamyQ::TAT-Ha::AmilCP)	1.1 μL (dilute ⅛)
Insert Terminator (B0015)	1.3 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL
Sterile water	10.0 μL

T4 DNA ligase	1 μL
BsaI-HFv2	1 μL
Total	20 μL

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

E: Spacer-Spacer-Spacer-Pveg-EEBD-RBS0-Csn::AmilCP-B0015

1. prepare the reaction as follow:

Vector (shuttle vector): pSTK-EXP-2-sfGFP	1.6 (dilute ⅕)
Insert transcription unit 1(Spacer A): pSTK-1A-Spacer	1.9 μL (dilute 1/10)
Insert transcription unit 2(Spacer B): pSTK-1B-Spacer	1.7 μL (dilute 1/10)
Insert transcription unit 3(Spacer C): pSTK-1C-Spacer	1.6μL (dilute 1/10)
Insert transcription unit 4(Pveg-EEBD-RBS0-Csn::AmilCP-B0015)	2.4 μL (dilute ⅕)
T4 DNA ligase buffer	2 μL

Sterile water	6.9 µL
T4 DNA ligase	1 µL
BsmbI	1 µL
Total	20 µL

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsmbI restriction enzyme)	10 minutes	1 cycle

F: Pveg-EEBD-RBS0-GanA-Gala-B0015

1. prepare the reaction as follow:

Vector (shuttle vector): pSTK-EXP-1-sfGFP ● Concentration (ng/µL): 162 ● bp: 6929	1.4 µL (dilute 1/5)
Insert promoter(Pveg) ● Concentration (ng/µL): 280 ● bp: 2361	1.1 µL (dilute 1/10)
Insert RBS(EEBD-RBS0) ● Concentration (ng/µL): 188 ● bp: 2113	2.1 µL(dilute 1/10)

Insert CDS (GanA-Gala) ● concentration(ng/μL): 177 ● bp: 2080	1.5 μL (dilute 1/10)
Insert Terminator (B0015) ● concentration(ng/μL): 107 ● bp: 2213	1.3 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL
Sterile water	8.6 μL
T4 DNA ligase	1 μL
BsaI-HFv2	1 μL
Total	20 μL

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

2. Pmol(0.05) insert: vector=1:1

D: Pveg-EEBD-RBS0-SamyQ::TAT-Ha::AmilCP-B0015

1. prepare the reaction as follow:

Vector (shuttle vector): pSTK-EXP-1-sfGFP (bp:6929)	1.4 µL
Insert promoter(Pveg)	1.1 (dilute 1/4) (1 µL insert in 3 µL water)
Insert RBS(EEBD-RBS0)	1.1 (dilute 1/3) (1 µL insert in 2 µL water)
Insert CDS(SamyQ::TAT-Ha::AmilCP)	1.1 (dilute 1/2) (1 µL insert in 1µL water)
Insert Terminator (B0015)	1.3 (dilute 1/2) (1 µL insert in 1 µL water)
T4 DNA ligase buffer	2 µL
Sterile water	9.94981071 µL
T4 DNA ligase	1 µL
BsaI-HFv2	1 µL
Total	20 µL

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

E: Spacer-Spacer-Spacer-Pveg-EEBD-RBS0-Csn::AmilCP-B0015

1. prepare the reaction as follow:

Vector (shuttle vector): pSTK-EXP-2-sfGFP	1.6 μL
Insert transcription unit 1(Spacer A): pSTK-1A-Spacer	1.4 μL(dilute 1/3) (1 μL insert in 2 μL water)
Insert transcription unit 2(Spacer B): pSTK-1B-Spacer	1.2 μL(dilute 1/3) (1 μL insert in 2 μL water)
Insert transcription unit 3(Spacer C): pSTK-1C-Spacer	1.2 μL(dilute 1/3) (1 μL insert in 2 μL water)
Insert transcription unit 4(Pveg-EEBD-RBS0-Csn::AmilCP-B0015)	1.2 μL
T4 DNA ligase buffer	2 μL
Sterile water	9.4μL
T4 DNA ligase	1 μL
BsmbI	1 μL
Total	20 μL

3. Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle

60 °C (Heat inactivation of BsmbI restriction enzyme)	10 minutes	1 cycle

F: Pveg-EEBD-RBS0-GanA-Gala-B0015

1. Prepare the reaction as follows:

Vector (shuttle vector): pSTK-EXP-1-sfGFP ● concentration(ng/μL): 162 ● bp: 6929	1.4 μL
Insert promoter(Pveg) ● concentration(ng/μL): 280 ● bp: 2361	1.1 (dilute 1/4) (1 μL insert in 3 μL water)
Insert RBS(EEBD-RBS0) ● concentration(ng/μL): 188 ● bp: 2113	1.1 (dilute 1/3) (1 μL insert in 2 μL water)
Insert CDS(GanA-Gala) ● concentration(ng/μL): 177 ● bp: 2080	1.1 (dilute 1/3) (1μL insert in 2 μL water)
Insert Terminator (B0015) ● concentration(ng/μL): 107 ● bp: 2213	1.3 (dilute 1/2) (1 μL insert in 1 μL water)
T4 DNA ligase buffer	2 μL
Sterile water	10.0 μL
T4 DNA ligase	1 μL
BsaI-HFv2	1 μL
Total	20 μL

2. Use the thermocycler to run the program below:

Temperature	Time	Cycles

37 °C (Restriction enzyme digestion)	1.5 minutes	25-45 cycles
16 °C (Ligation)	3 minutes	
37 °C (Restriction enzyme digestion)	5 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	10 minutes	1 cycle

Done by Abigail, Abby T, and Chloe

Date: 20260629
Title: 500mL agar plate
Purpose: prepare arabidopsis for vertical plate testing
Procedure:

1. Prepare the following

1/2 MS medium powder (with sucrose)	1.15g
MES buffer	0.25g
ddH_2O	500mL
sucrose(1%)	5g
Agar powder(1%)-add last	5g

2. Add contents except agar powder into 500mL flask
3. Stir with hot plate & magnet until clear of sediments
4. Use pH probe to measure pH of the solution(pH 5.7~5.8)
    - Add 0.1M KOH if pH level is low and until it reaches the desired pH value
5. Distribute to 2 500mL flasks
6. Add 2.5g agar powder each
7. Stir until no sediments
8. Wrap necks with aluminum foil and stick an indicator tape on top
9. Autoclave for 20 minutes at 127°C
10. Transfer flasks to BSC
    - Store in hot bath if not immediately pared
11. Dry clean 13sec surface with wipes 10% FtH
12. Pour 50mL of agar solution per plate

Done by Ethan C
Supervised by Gabriel

Date: 20260629

Title: Miniprep
Purpose: isolate DNA for exemplification (F2&F8)
Procedure:

1. Transfer 1.8mL TB medium to a 2mL microtube
2. Centrifuge at 16000 g for 1 min
3. Discard the supernatant
4. Repeat steps 1-3 until all of the medium is collected.
5. Add 200µL of MX1 buffer
6. Resuspend pellet via pipetting/vortexing
7. Add 250 µl of MX2 buffer
8. Gently mix via inversion 8-10 times
9. Incubate at room temperature for 2 min
10. Add 350µL MX3 buffer
11. Immediately invert 4-5 time
12. Centrifuge at max rcf(23444 g) for 5 min
13. Assemble column & collection tube in the meanwhile
14. Transfer supernatant(800µL) with 200µL micropipette to avoid contaminating the pellet
15. Centrifuge 7kg for 30sec
16. Discard flow-through
17. Add 500µL of WN buffer
18. Centrifuge 7000 g for 30sec
19. Discard flow-through
20. Add 700µL of WS buffer
21. Centrifuge 7000 g for 30sec
22. Discard flow-through
23. Centrifuge empty column 10000 g for 3 min
24. Place column to 1.5mL microtube
25. Add 50µL of EB buffer directly on the membrane
26. Incubate column at room temperature for 3min
27. Centrifuge 10000 g for 3 min

Done by EthanC
Supervised by Gabriel

Date: 20260630
Title: LB Agar plate
Purpose: prepare LB agar plate
Procedure:

1. Measure 7.5 g of LB broth powder and 4.44g of agar powder in 300 ml of distilled water.
2. Use a hot plate and stirring rod to stir until no sediments remain in the solution.
3. Autoclave at 127°C for 20 mins.
4. Bring the solution to the BSC and add 300 µl of Chloramphenicol into the solution
5. Pipette the solution 20 ml into each of the petri dishes and leave to dry.

Done by Ryan Wei and Naomi
Recorded by Ryan Wei.

Date: 20260630
Title: Golden Gate Assembly
Purpose: To assemble Pveg-EEBD-RBS0-spacer-GanA-Gala-histag-B0015 (pmol = 0.05, vector:insert = 1:1)
Procedure:

1. Prepare the reaction as follows:

Vector pSTK-EXP-1	1.39 µL
Insert Pveg	1.10 µL (dilute 1/4)
Insert EEBD-RBS0	1.10 µL (dilute 1/3)
Insert spacer (STK244-pSTK-0Cx)	1.07 µL (dilute 1/2)
Insert GanA-Gala	1.15 µL (dilute 1/3)
Insert 6x His Tag	1.01 µL (dilute 1/2)
Insert B0015	1.34 µL (dilute 1/2)
T4 DNA ligase buffer	2 µL
Sterile water	7.84 µL
T4 DNA ligase	1 µL
BsaI-HFv2	1 µL
Total volume	20 µl

2. Use the thermocycler to run the program below

Temperature	Time (min)	cycles

37˚C (restriction enzyme digestion)	1.5	25-45
16˚C (ligation)	3	25-45
37˚C (restriction enzyme digestion)	5	1
60˚C (heat inactivation of BsaI)	10	1

Done by Naomi, recorded by Naomi.
Supervised by Gabriel

Date: 20260630
Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:

1. Turn on the water bath, set it to 42°C and prepare ice.
2. Mix 60 µl competent cells with 10 µl plasmid in each of 6 respective tubes.
3. Incubate on ice for 20 min.
4. Heat shock at 42°C for 45 sec.
5. Incubate on ice for 3 min.
6. Add 1 ml of SOC medium and incubate at 37°C for 1 hr.
7. Centrifuge at 5000 g for 5 min and remove 1 ml of the supernatant.
8. Resuspend the remaining mixture and spread the mixture onto the LBA plate.
9. Place the plates into the incubator after labelling.

Done by Ryan Wei, recorded by Ryan Wei.

Date: 20260630
Title: Liquid culture
Purpose: grow and amplify bacterial cells from single colonies
Procedure:

1. Label liquid culture tubes.
2. Turn on the bunsen burner.
3. Add 3 ml of LB medium and 3 µl of Chloramphenicol into each tube.
4. Dip a pipette tip into the colonies.
5. Dispense the tip into their respective tubes.
6. Place the tubes into the incubator at a slanted angle.

Done by Ryan Wei, recorded by Ryan Wei.

Date: 20260630
Title: 0.8% Agarose Gel Preparation for Gel electrophoresis
Purpose: Make 13 well agarose gel for gel electrophoresis
Procedure:

1. Prepare and assemble the mold and the comb.
2. Prepare 0.32 g of agarose powder and add it into a flask.
3. Add 40 mL of 1X TAE solution into the flask.

4. Cover the flask with plastic wrap, then use a tip to puncture 3 to 5 holes on the plastic wrap.
5. Microwave for 30 sec then agitate it. Repeat this step until there's no sediments in the melted gel.
6. Wait until the gel stops smoking but is still hot.
7. Add 1 µL of safety dye into the gel and agitate it.
8. Pour the melted gel into the tray.
9. Use a pipette tip to press out bubbles.
10. Leave to solidify at room temperature.

Done by Naomi, recorded by Naomi
Supervised by Gabriel